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Hazard Analysis Application
CMP Final Year Project

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1. Introduction

This section introduces the project, providing the background and motivation for the application's development and outlining the project's overall aims and objectives.

1.1. Background and Motivation

Hazard analysis is a critical process in projects that involve delivering physical products or new processes to customers. I studied this topic during my Systems Analysis module and applied it during my Year in Industry placement. The organisation that I was on placement at is an engineering company that had several projects in which physical pieces of hardware were delivered to an area of the business or a customer. Due to the nature of the projects being related to aerospace and defence, there was an increased effort to ensure all products are held to a standard of inspection and auditing, with guidance on how this is performed given by the Ministry of Defence (MOD). This is so that if there is a fault or incident, there is an audit trail which will support a thorough investigation. As a result, hazard analysis must be performed on each release or update to a product to ensure that risks are sufficiently mitigated. The way that hazard analysis is currently performed at the organisation is by a legacy Excel template, which relies heavily on embedded formulas to generate documentation for sign-off by the project's Design Authority (DA).

However, this approach presented several challenges:

- The spreadsheet was created years ago, and the formula logic is not well understood by current users (very little has been done in terms of development from the original).
- Knowledge transfer was not a priority within the organisation which has led to the current solution (Excel based) producing frequent errors and misuse.
- The Head of Safety noted difficulties in managing the volume of documents and inconsistencies caused by incorrect formula usage.

A request had previously been made for a custom application that would assist in creating and maintaining hazard logs. However, due to it not being seen as a priority, projects requiring hazard analysis are to remain being completed using the legacy Excel template. Under the guidance of the MOD (2018), the hazard log would normally be implemented as some form of computer database. However, the alternative for analysing low complexity systems, which have fewer hazards, is that it may be more suitable to maintain the data using an application like Word or Excel.

1.2. Project Aim

Therefore, to address these issues, my main aim is to develop a centralised Hazard Analysis Application using Oracle APEX (Application Express) – a web-based, low-code platform built on top of an Oracle database that specialises in data-orientated application development. The reason for Oracle APEX in particular is that it was a development tool I used frequently on placement and has been used to develop several internal tools used within the organisation.

This application will:

- Provide a standardised template for all projects hazard analysis to follow.
- Serve as a single source of truth for hazard analysis data.
- Enable a streamlined review and approval process, including automated document generation for later DA sign-off.
- Look to replace Excel spreadsheets with structured database tables, improving data integrity and accessibility.
- Utilise Oracle APEX's reporting and querying capabilities to present and manipulate data through an intuitive front-end interface.
- Allow for implementation of additional, more complex components if required

There are already several developers working within my placement organisation that are experienced with Oracle APEX. Thus, if the application was adopted, there would be sufficient resources and knowledge to provide in-service support to such an application in the future. The benefits of this application as opposed to the existing legacy Excel template is that any error-prone Excel formulas are removed. Due to there being limited knowledge on Excel formulas, removing dependencies (such as the reliance on those with knowledge) would be beneficial for long-term usability. Additionally, having a single source of truth as a repository for projects means consistency of that data is improved, allowing for traceability across all projects. This supports collaboration across teams and the safety authorities as the administrative overhead and document management problems are reduced.

Alongside the application will be a set of supporting documents captured as sections within this Project Portfolio document. This list is from the minimum documentation set required for applications within the organisation and provides an outline for the section headings in this document to enable the same structure on the project.

- Project Management Plan – outlines how the project will be managed and the tools used
- System Requirements Specification – Functional and Non-Functional Requirements
- System Design Document – detailing system architecture
- Test Plan – testing strategy

- Test Results – results from running test cases as well as from user testing
- Software Release Note – summarising features for deployment ***(this was not seen as essential as part of the project portfolio and has not been included)***

1.3. Objectives

It is important at the early stages of the project to define the key objectives for them to be reflected upon at the end as a measure of its success. These objectives will inform the decisions that are made regarding the project, including planning, design and implementation.

- To perform academic research into the chosen subject area and surrounding tools/processes concerning hazard analysis and application development.
- To create a set of Functional and Non-Functional requirements that are to be used to inform the development of the system.
- To produce a comprehensive set of technical documentation in line with my placement organisation. This includes Project Management Plan, System Requirements Specification, System Design Document, Test Plan and Test Results.
- To produce a functioning Oracle APEX application (along with a suitable database structure) which fits the purpose of hazard analysis.
- To develop sufficiently complex features that can be incorporated within an Oracle APEX Application and that meet the project requirements for CMP accreditation.
- To perform thorough and robust "formal testing" which includes a combination of both developer and user testing.
- To seek guidance and feedback from developers that use Oracle APEX from my previous placement to ensure the system is well built and to an industry standard.

2. Literature Review

The aim of this literature review is to gain an understanding of identified literature related to various topics within this research area. For the purposes of this project, standards for hazard analysis, existing tools for performing hazard analysis, how Oracle APEX can be used to develop applications, and other aspects of the topic area that can be incorporated in the project have been researched and the relevant literature has been documented in this section.

2.1. Hazard Analysis

2.1.1. What is Hazard Analysis

Hazard analysis is an important process in project development that identifies and evaluates potential risks associated with a product or process, particularly when the product is to be deployed or given to customers. Its primary goal is to ensure safety and compliance by identifying and controlling hazards before they result in an accident. EASE (2009) provides the guidance on hazard identification and how it fits into the wider hazard analysis process.

A complete guide to hazard analysis, including all the specific processes, has been produced by ASEMS (Acquisition Safety and Environmental Management System) in their Part 2 Guidance - POSMS (Project Oriented Safety Management System) (2016). The methodology that has been established in this guidance form the fundamentals for hazard analysis on engineering projects, especially those in the defence sector (ASEMS, 2016). As this is already common practice for my placement organisation, it would be reasonable to use this way of working in the application.

2.1.2. Defining Risk

Risk is defined by EASE (2009) as *"the combination of the predicted frequency and severity of the consequences of hazard(s) taking into account all of the potential outcomes".* All hazards need to be assessed against risk as it can be used to determine if a hazard is at an acceptable risk level or As Low As Reasonably Practicable (ALARP). ALARP requires that risk be reduced to the lowest level that is reasonably achievable considering project constraints such as cost, effort and resources. Figure 1 demonstrates what combination of severity and probability would be accepted as ALARP.

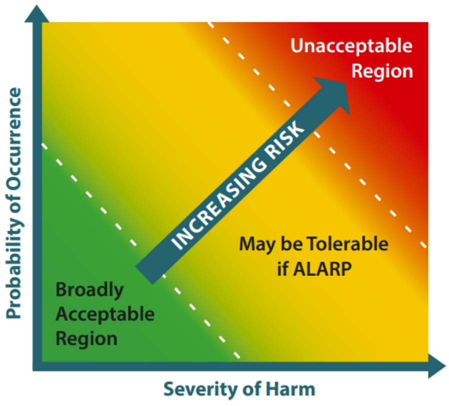

Figure 1 Risk as a Combination of Severity and Probability (MOD, 2020)

2.1.3. Hazard Logs

Importantly, the identified risks are documented in hazard logs. As highlighted by Cutajar (2023), hazard logs play a key role in the hazard analysis process by systematically documenting each identified hazard, its associated risk level, and the rationale for whether it is deemed acceptable or is considered ALARP. It is an entity that must be continuously monitored and updated to reflect any new hazards, changes to the system, or updates to the operational requirements, requiring a new iteration after every change. The Safety Artisan (2023) highlights that an advantage of using a hazard log is that it provides full traceability of how safety has been managed during the lifecycle of a project. This is essential for regulatory compliance, auditing and accountability. Hazards that are no longer relevant to a particular project are not deleted but instead marked "obsolete" or "out of scope" for historical data to be reviewed if needed, maintaining the need of full visibility for auditing purposes.

2.2. Existing Hazard Analysis Applications

Having outlined hazard analysis, it is now important to introduce existing applications that are commercially available as well as other alternatives that perform the same functionalities. A motivation for this project is that the organisation where I was on placement was using a legacy Excel template. The problems with this approach have already been discussed in Section 1.1, however they are reinforced by MOD (2018). It is acknowledged that performing hazard analysis in applications like Excel make producing a final document easier. However, it is the experience, knowledge of specific Excel functionalities and tools of the author that produces a well written document.

On the other hand, there are existing hazard log tools that provide functionality for connecting entities and information in the hazard analysis process within the engineering sector (Cutajar, 2023; The Safety Artisan, 2023). Notably, the use of these tools must be applied in a way that will be effective in supporting the risk management process. The following sections provide an overview of such tools, which can be used later in the design stage of this project to help better inform processes and presentation of the application.

2.2.1. eCassandra

First, eCassandra was being considered for adoption by the organisation that I was on placement with as it is seen as a viable tool within the industry. eCassandra is one of the *"MOD-*

preferred tools for constructing hazard logs" and has been mandated for all new Defence Equipment and Support (DE&S) projects (MOD, 2020). ASEMS (2016) defines all the components that are captured within the eCassandra hazard log as: Hazard, Cause, Accident, Control and Reference which are the same entities used in the legacy Excel template. The relationships between the components are shown by the Figure 2.

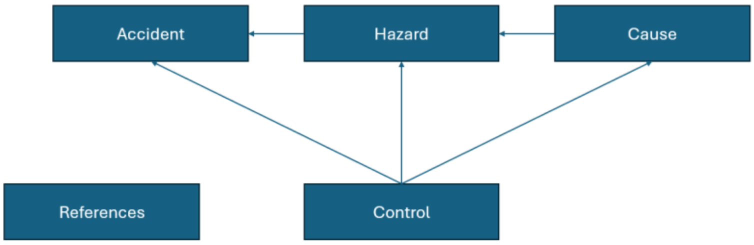

Figure 2 Hazard Log Entity Relationships

Access to each project can be controlled via user permissions within the application through "roles" within a project. This ensures that the appropriate authority is creating and reviewing the correct data in line with organisation policy. Once a project has been created, the users can start populating the hazard log. Having a page such as Figure 3 allows for user inputs to be regulated so they are only providing valid information, something which would be harder to maintain and not completely preventable if an Excel spreadsheet was used for this stage.

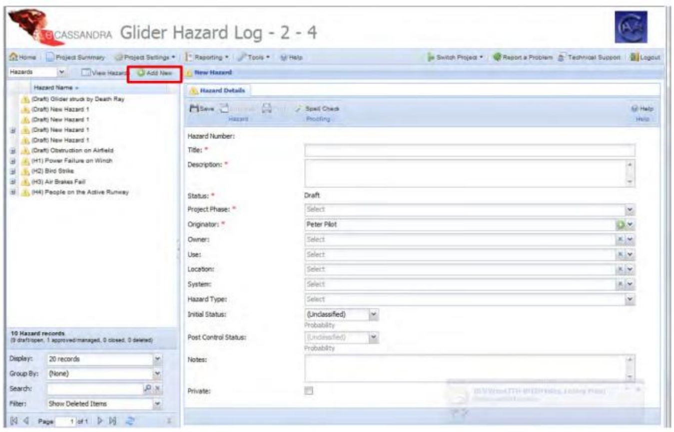

Figure 3 eCassandra New Hazard

Once all entities have been identified, they can be linked together to create the fully defined hazard log. There are five types of linkages between entities that are required: Hazard – Cause, Hazard – Accident, Hazard – Reference, Cause – Control, and Accident – Control. eCassandra will allow other linkages to be added, but the five listed are sufficient to model the hazard log based on the structure adopted by eCassandra ASEMS (2016). The advantages of using a solution like this is that it is highly scalable and can be used across multiple projects, providing a central repository for the Safety Authorities to manage all projects in one place. On the other hand, there is a considerably greater level of overhead which may not be suitable for all organisations as maintaining a no-SQL database would require additional expert knowledge of its entities and relations.

2.2.2. ISCaDE™ Pro

Another tool is ISCaDE™ Pro, which develops safety cases using a standardised approach, working alongside a DOORS database which is an existing tool that has been adopted by professional organisations to manage requirements (ISCaDE, 2014). It can be applied to all types of safety critical systems and provides multi-user, multi-access, object-oriented and graphical presentation capabilities together in an integrated environment. ISCaDE™ Pro was initially developed for the rail industry to improve customer service (ISCaDE, 2014). However, because the architecture of ISCaDE™ Pro is flexible and dynamic, it can be used to fulfil the requirements of other safety critical systems.

Considering the aforementioned risk information highlighted by MOD (2020), ISCaDE™ Pro captures this information and then utilises it in the following ways: risk matrix (before mitigation of the risk), risk matrix (after mitigation of the risk), displaying any one or both of the risk matrices, and displaying number of objects lying in a specific risk class category as well as the risk class name. After defining objects via forms, they can be linked together through a User Interface (UI) seen in Figure 4, which cannot be achieved in Excel.

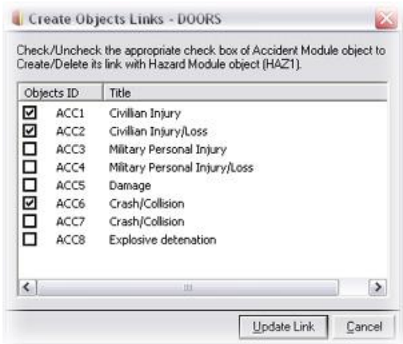

Figure 4 Link Creation form ISCaDE™ Pro

2.3. Oracle APEX

Finally, Oracle APEX (Application Express) is a web-based, low-code platform built on top of an Oracle database. Accessible through a web browser, APEX enables developers to rapidly build scalable and secure applications without needing extensive programming knowledge

(Kvet, 2024). Since APEX is integrated with a database, all application logic, data access, and page processing are executed within the schema environment using SQL and PL/SQL. Importantly, the platform features a visual, drag-and-drop interface that simplifies the creation of components such as forms, charts, reports, and dashboards. This approach makes APEX well suited for data-driven applications, where rapid development and database interaction are important. Additional information about APEXs functionalities and capabilities is provided in Oracle APEX Documentation (Oracle APEX, 2025).

2.3.1. Rapid Application Development with Oracle APEX

With the ever-present need for organisations and their data to move online, there is an increasing trend for cheap and reliable IT solutions. Oracle APEX offers the ability for applications to be defined and deployed without the proper knowledge of programming languages or the supporting technologies. This is one of the main appeals and advantages of low/no-code tools as explored by Kvet (2024) who makes a link between the emerging application development tools, such as Oracle APEX, and the industries increased complexity but lack of demand for specialised developers. They go on to state that because of this, developers in more complex fields have become specialised whereas nontechnical staff are able to perform in developer roles.

2.4. Use of AI in Hazard Analysis

In addition to these tools, AI is becoming increasingly used in every aspect of project development, so it is of interest to explore where and how the use of AI can be implemented to perform hazard analysis. There have been investigations to how Larg Language Models (LLMs) can be used to enhance the hazard analysis process, such as one performed by Diemert and Weber (2023) who mainly explored OpenAIs Chat-GPT3. The purpose of the study was to show an LLM's ability to suggest alternative ideas and lines of investigation to be performed by a human analyst. This provided a second source of ideas to perform a "Co-Hazard Analysis". In conclusion, Diemert and Weber (2023) did not provide a definitive list of benefits for the use of LLMs. However, they do note that the AI would produce valid points of discussion, but only a low number (around 35%) of words were deemed *"correct and useful"*.

A similar investigation was also conducted by Qi, Zhao, Khastgir and Huang (2025) using Chat-GPT3.5. The main outcomes from their investigation were that the application of

ChatGPT without the knowledge of a Subject Matter Expert (SME) would produce unreliable outcomes and notes that the LLM would tend to produce more conservative and less comprehensive responses. A reasoning for this is that LLMs require a large amount of high-quality data to be trained effectively. However, due to the nature of safety critical projects, there may not be enough data that can be disclosed for the purpose of training. This may lead to LLMs only being able to output general information regarding a project which is less useful.

2.5. Summary

In summary, Oracle APEX has been proven as a suitable tool for the rapid development of data-driven applications supported by Kvet's (2024) discussions on its suitability for data management as well as low requirement of technical knowledge from developers. Developing in such a way would work best with an agile environment, prioritising the iteration of development to create slight but consistent improvements to the application. The research into hazard logs, the data that drives them as well as the existing applications, and those recommended by the MOD (2020), has shown that representing hazard log data in the form of an Oracle APEX application would be feasible. Supporting documentation would be created in line with findings from Islam, Hasan and Eisty (2023) due to the project being managed in an agile way. With regards to the use of AI for hazard analysis, it is noted by Diemert and Weber (2023) that an LLM's potential is largely unexplored and that a main concern from Qi, Zhao, Khastgir and Huang (2025) is the trustworthiness of the responses produced.

3. Project Management Plan and Methodology

3.1. Purpose

The purpose of this section is to outline the aims, objectives and ways of working in the project to ensure that they are documented and followed throughout. This aids in the development process as a reference point, giving a full overview of the project and its components.

When choosing how this project was to be completed, a common practice for industry software development is an agile way of working. Adopting this approach suits the type of iterative development proposed, particularly when using rapid application development tools such as Oracle APEX (Kvet, 2024).

3.2. Existing vs Proposed Solution

Prior to development, it is valuable to analyse the existing process (initially discussed in Section 1.1) in more detail to get a better understanding of the benefits of changing to a new system. Additionally, a representation of what the proposed solution will look like with the use of the application has been created and is demonstrated in Figure 5.

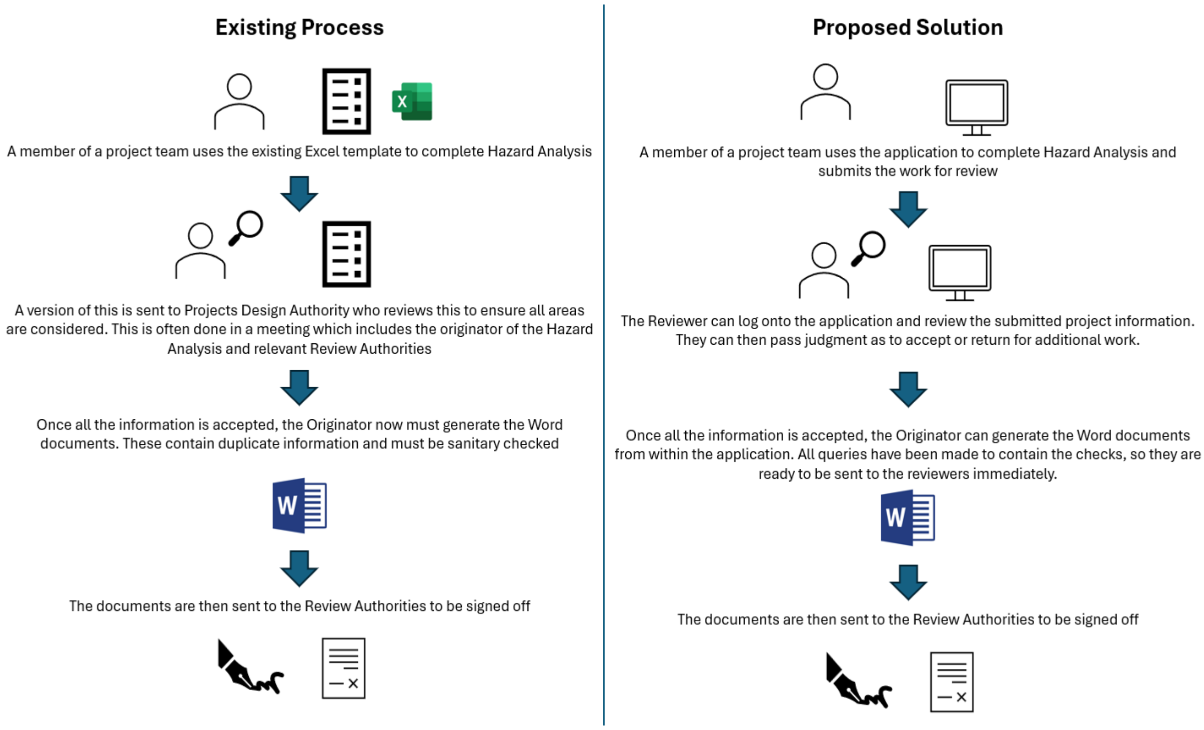

Figure 5 Existing vs Proposed Process

3.3. Gap Analysis

At the early stages of the project, it is important to identify the differences between how a system currently operates and how a desired future state would work. As part of the Gap analysis on this project, alternatives of existing applications have been included to accurately assess whether this project proposes a unique solution that isn't already available. The chosen existing applications are eCassandra and ISCaDE™ Pro which have already been discussed as part of Section 2.2. The full Gap Analysis can be found in Appendix D (Gap Analysis).

3.4. Tools Used

This is to provide a complete list of all the tools that have been used throughout the project.

Tool Used	Description
Oracle APEX	Online application used for the development of the application described in the following sections
Trello	Online application used to plan and track project tasks
Jira	Online application used for the tracking of requirements and test cases
GitHub	Online application used for the version controller for code on the project - https://github.com/TB52052/CMP-Third-Year-Project
Diagrams.net	Online application used for the creation of diagrams used for documentation of the system
Spyder 6	Python IDE used to develop python code
Visual Studio Code	IDE used to develop .js code
Terminal	Local Terminal used to run python and .js code

Table 1 Tools used on the Project

3.5. Ethics

As there will be interaction with people in the form of feedback and user testing, in line with the UEAs' guidelines on managing personal data on research projects, an Ethics application (ETH2526-0560) was made and approved by the SCI S-REC (Faculty of Science Research Ethics Subcommittee) prior to any communications with testers and form of data collection.

4. System Requirements Specification

4.1. Purpose

The purpose of this system requirements specification is to provide a detailed guide for the project, ensuring that the application is built to meet all needs previously identified. Each need has been translated into a set of requirements that will then be linked to test cases where formal testing will provide evidence that they have been met and are compliant.

4.2. Managing Requirements

MoSCoW (Must, Should, Could and Won't) analysis allows those writing the requirements to determine the priority of features of a proposed system and how important they are to include in the development stage (Sommerville, 2016). This has been done for all the following requirements to provide better insight into what features are essential and need to be implemented in the first iteration of the application. Requirements help to create appropriate test cases when testing the solution. Due to this, it is important to have a centralised repository of requirements to ensure a single source of truth and so that they can be linked to test cases in the future. As a result of this, Jira has been chosen as the tool to store the requirements.

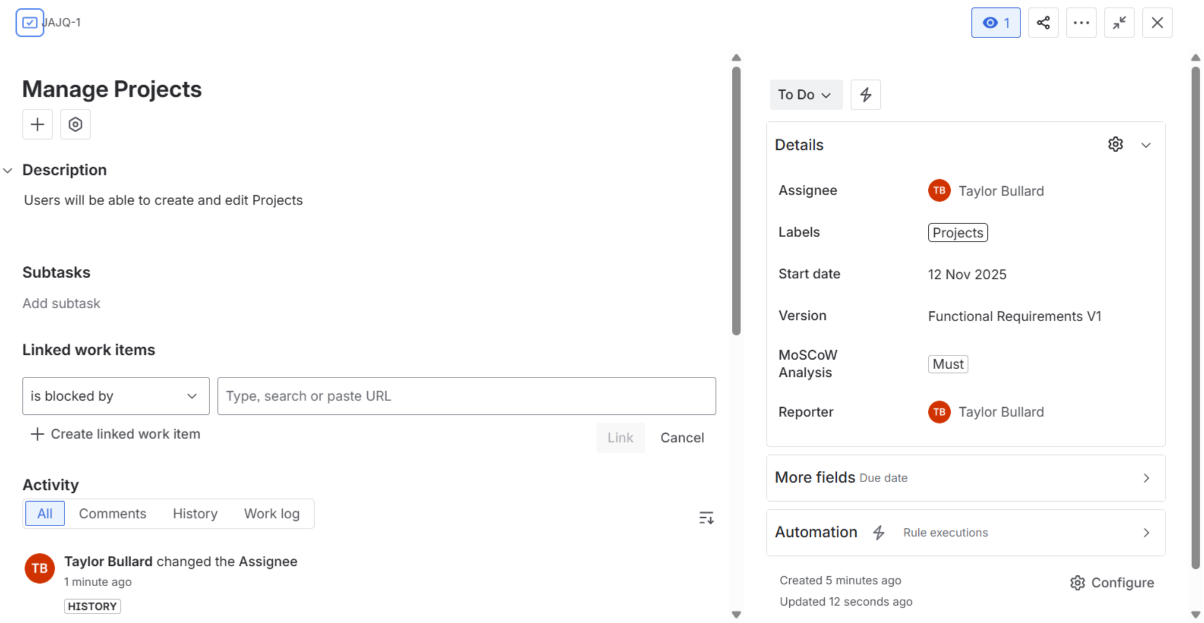

Figure 6 Requirements in JIRA

Figure 6 shows how requirements are written in JIRA. Additional fields such as Version and MoSCoW Analysis have been created to capture additional pieces of information. The tags section was used to create Epics. This will allow test cases to be assigned to a group of requirements rather than having to assign them individually, ensuring that all individual requirements are compliant. Each individual requirement uses the parent/child feature to be assigned to the epic as shown in Figure 7.

Figure 7 Linking Requirements in JIRA

4.3. Functional Requirements

This section lists the Functional requirements for the system. Each of the requirements listed has been assigned a unique ID to allow for full traceability across the lifecycle of the project. Functional requirements can be found in Appendix A (Functional Requirements) with a sample shown in Table 2 (Requirement ID, Requirement Name, Description, Tag, Version, MoSCoW).

Requirement ID	Requirement Name	Description	Tag	Version	MoSCoW
JAJQ-1	Manage Projects	Users will be able to create and edit Projects	Projects	Functional_Requirements_V1	Must
JAJQ-17	Project Dashboard	All projects will be displayed on a Project Dashboard	Projects	Functional_Requirements_V1	Must
JAJQ-37	Reviewer Dashboard	Reviewers will have a dashboard showing pending review items	Dashboard	Functional_Requirements_V1	Could

Table 2 Sample of Functional Requirements.

4.4. Non-Functional Requirements

This section lists the non-functional requirements for the system which can be found in Appendix B (Non-Functional Requirements). It is important to note that as Oracle APEX is being used to develop the application, there are certain constraints and existing non-functional requirements that are managed as part of the APEX implementation. For example, user accounts are a functionality already provided by APEX that will be used for this application. Additionally, there aren’t as many security requirements as if the application was developed in an alternative way due to this being basically being implemented by APEX. There is the ability to design a custom security system using the database however this is currently not a need for the application at this stage due to the main purpose being showing hazard analysis capabilities.

5. System Design

5.1. Purpose

The purpose of this System Design (SD) section is to document design decisions and describe how the Functional and Non-Functional requirements (Section 4) transform into specific design specifications. This SD section describes considerations and provides a high-level overview of the system architecture. The high-level system design is further decomposed into low-level detailed design specifications for certain components. Typically, this section of system design would be given to the development team to provide instructions on the system architecture. Design documents are iteratively worked on during the system development life cycle (SDLC) so will be updated with the latest progression development.

5.2. Oracle APEX

As previously mentioned, Oracle APEX is staged on top of an Oracle Database. Figure 8 demonstrates how the user accesses this through the front-end webpage, connecting via ORDs.

Figure 8 Oracle APEX - User Request Process (Oracle - https://apex.oracle.com/en/platform/architecture/)

5.3. Table Definition

A definition of all database tables has been made to provide full traceability in case of any future updates or iterations to the application and can be found in Appendix H (Table Definition). Additionally, an ER Diagram providing a visual representation of the database structure can be found in Appendix C (ER Diagram).

With all the tables, auditing columns have been included. CREATED_BY, CREATED_DATE, UPDATED_BY, and UPDATED_DATE are used to keep a record of the user that has made amendments to a data entry. As well as this, some tables (such as H_ACCIDENT_DATA, H_CAUSE, H_CONTROL, and H_HAZARD) include a LOGICALLY_DELETED column which means entries will not be permanently deleted from the system. This is beneficial for auditing and data integrity as it may still be valuable for users to see previously deleted entries as they could still be relevant to the project.

In the linking tables, a synthetic primary key (PK) has been used rather than a composite key between the two foreign keys (FKs). This is due to previous experience in using Oracle APEX where bugs would arise when trying to manually insert rows to an Interactive Grid when using data with composite keys. In addition to using a Synthetic PK, both FKs have a UNIQUE constraint ensuring that there is no duplication of data that can happen.

5.4. Processes

The following section demonstrates some of the important processes of the application as defined by the "Must" criteria of Functional Requirements. Alongside explaining the functionality, extracts of code/processes have been included should these components require future development or in-service support. The following figures are an initial design of the application which could change depending on later feedback and user testing.

5.4.1. Creating a Project

Users can navigate to the Project Dashboard page (Figure 9). This shows all Projects that are available to teams that the user is a part of and shows the status of associated tabs.

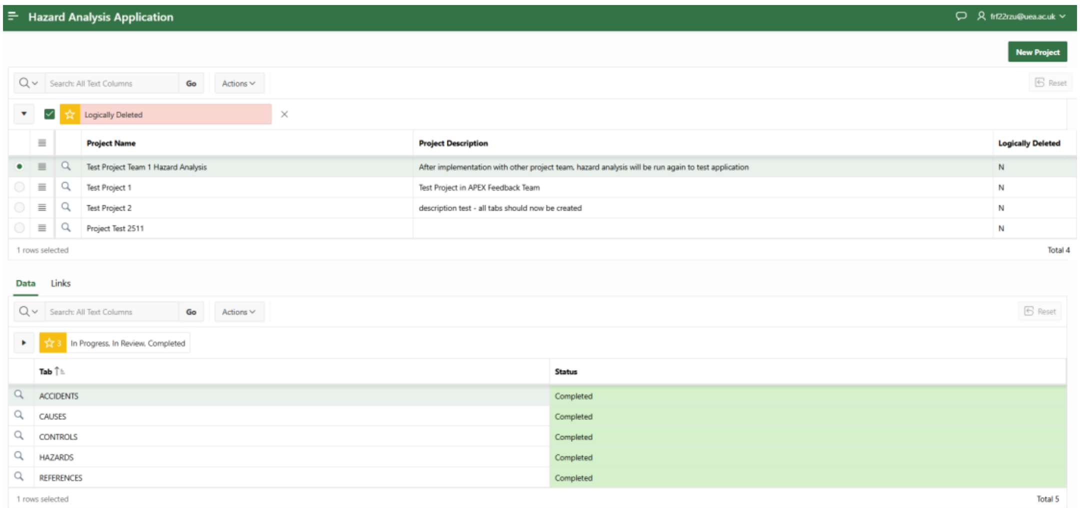

Figure 9 Project Dashboard Landing Page

From this Dashboard page, a user can use the "New Project" button to create a new project, taking them to a separate page to fill out more detailed information regarding the project. If the user needs to change any information after creating a project, they can access this through the buttons on the Project Detail which will then navigate to the same New Project Page, but in an editing mode (Figure 10).

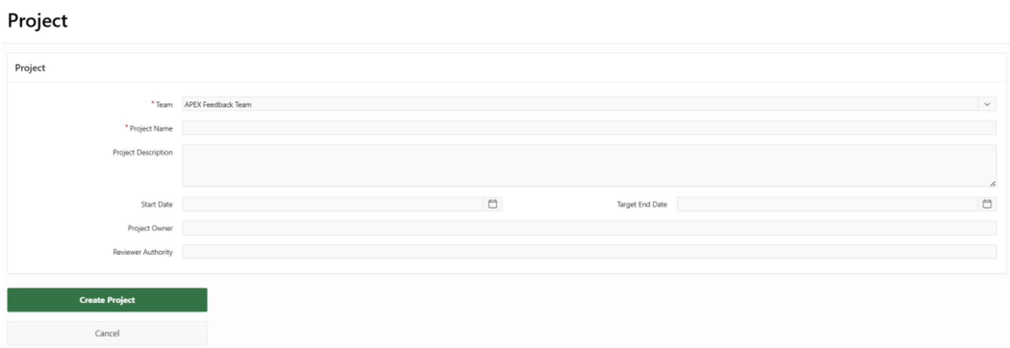

Figure 10 New Project Page

5.4.2. Page Navigation

As there are many possible routes away from the Dashboard page via the Interactive Grid, there needed to be logic set up to ensure the link column navigated to the correct page. A limitation of Oracle APEX is that multiple pages cannot be defined for a link column, therefore the work around for this was passing the tab type via &TAB_TYPE to a temporary page with branches that trigger on a server-side condition to direct to the correct page. A main drawback from this approach is computational time, however, this has been captured as a non-functional requirement that has its own specific test case to ensure that it is compliant and the page loads within a reasonable time. To better visualise this implementation, the following diagram represents the flow of where a user will be navigated to from this page (Figure 11).

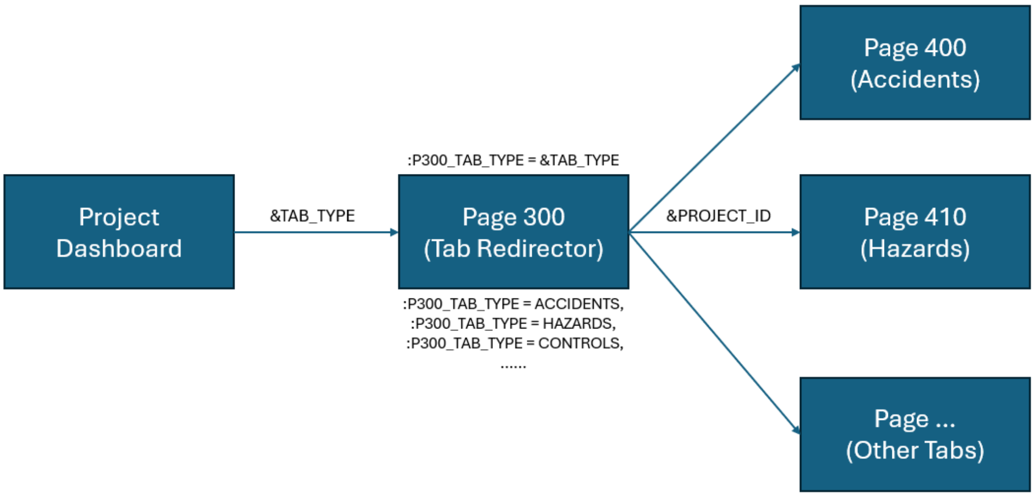

Figure 11 Tab Redirecting - Data Flow Diagram

5.4.3. Inputting Data

A benefit of Oracle APEX is its ability to present and make changes to data in a seamless way by using Interactive Grids, which are an out of the box feature. Interactive Grids (Figure 12) allow for "Excel like" freedom of editing as you can edit a row by just clicking on it. This was the reason for choosing this component feature as it is intuitive and has similar properties to Excel which the targeted users of this solution already have experience with. Additionally, Interactive Grids allow for a user to customise the data they see by hiding columns or ordering data based on their preferences i.e. Risk Level rather than the self-incrementing Code column. These were created for each Data tab (Accidents, Hazards, Causes, Controls and References) and can be navigated to on each using the tab navigation above the Interactive Grid.

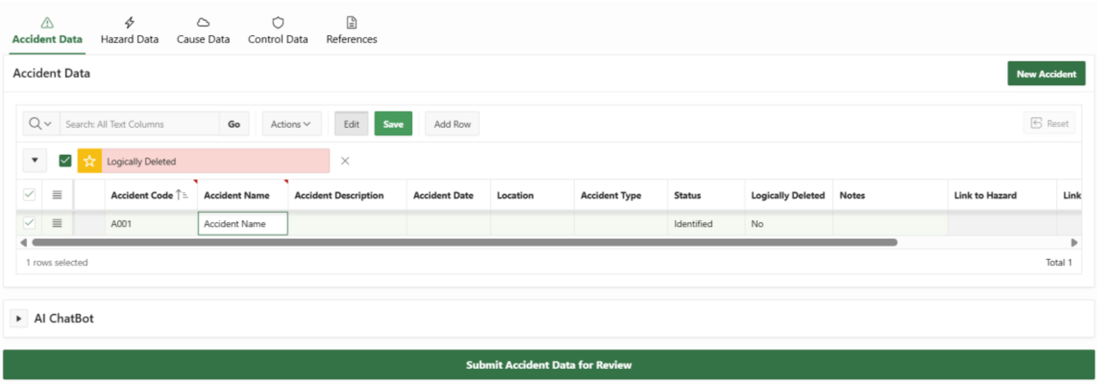

Figure 12 Data Interactive Grid

Alternatively, a modal page (Figure 13) can be used to both create and edit rows.

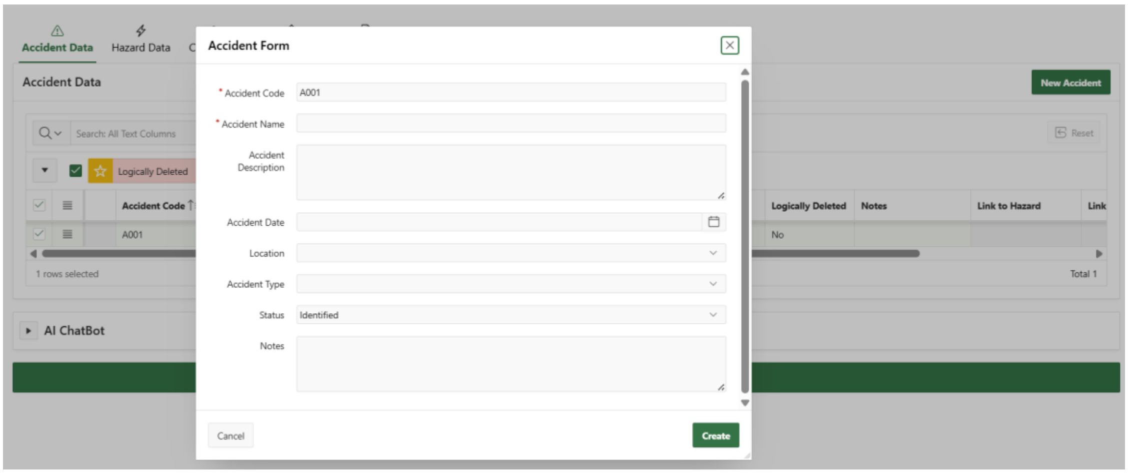

Figure 13 Data Modal Page

For hazards, an automatically updating field which calculates the overall Hazard Risk based on the Severity and Likelihood was implemented (Figure 14). The PL/SQL code below shows how the risk level is classified by multiplying the likelihood and severity values.

```
DECLARE
    lv_risk NUMBER;
BEGIN

    IF :P411_INITIAL_LIKELIHOOD IS NOT NULL
       AND :P411_INITIAL_SEVERITY IS NOT NULL THEN

        lv_risk := :P411_INITIAL_LIKELIHOOD + :P411_INITIAL_SEVERITY;

        IF lv_risk <= 3 THEN
            RETURN 1;
        ELSIF lv_risk = 4 THEN
            RETURN 2;
        ELSIF lv_risk <= 6 THEN
            RETURN 3;
        ELSIF lv_risk <= 8 THEN
            RETURN 4;
        ELSE
            RETURN 5;
        END IF;
    END IF;

    RETURN NULL;
END;
```

Figure 14 Risk Classification PL/SQL code

5.4.4. Linking Data

When linking the data, the decision was made to only make linking possible one way, i.e. Hazards can be assigned to a single Accident entry, but a single Hazard entry cannot be assigned to Accidents. This was done due to the effort required to create a unique page and table each time. User Feedback will be gathered to see if this decision affects the overall user experience and can always be added in future iterations of the application.

The decision to include the ability to link in this way (see Figure 15) was made due to ISCaDE™ Pro having a similar functionality (Figure 4). After mapping how a user navigates the application, having a modal page to link to other Data tabs would reduce to need for the user to go through different menu screens to find the page they are looking for. Additionally, if a user must go between screens frequently, this method proves to be much more efficient.

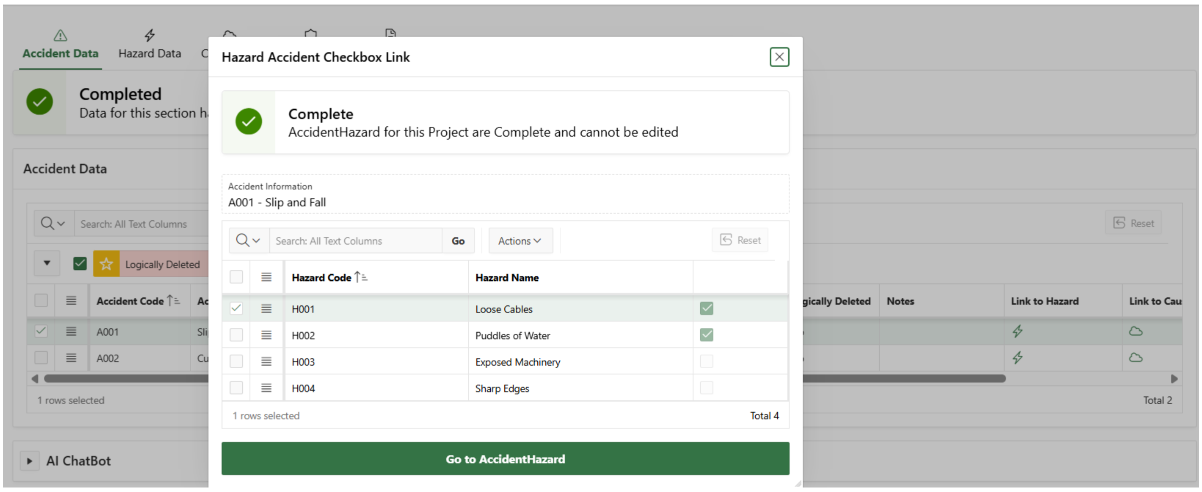

Figure 15 Link Modal Page

Furthermore, to view all the links in one place, a separate "Link" page (Figure 16) was created for each of the possible ways the data can link to each other. Entries can be made on this page using an LOV containing all the Data Tabs and has the ability to open a modal page for more information regarding the data should the user need to view it.

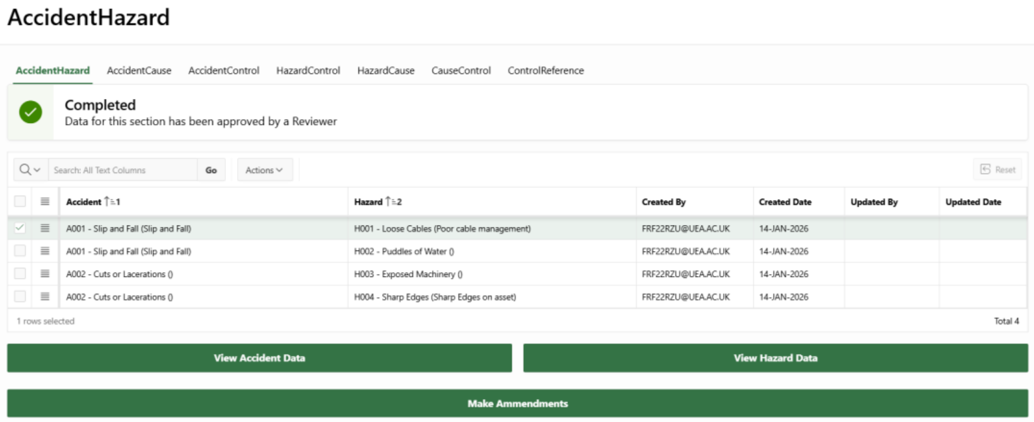

Figure 16 Link Page

5.4.5. Reviewing a Tab

A key requirement of this project was to include a "Review Loop" which allowed for users to submit the data for a Project and then for a reviewer to have to ability to view the submitted data and decide whether to then "approve" or "reject" the submitted information. The following figures demonstrate the process which has been applied to all Data and Link pages.

On the creation of a Project, all Tabs are assigned the status of "In Progress". Once a user has completed the information for a given Tab, they can submit the information to a teviewer via the "Submit for Review" button (Figure 17).

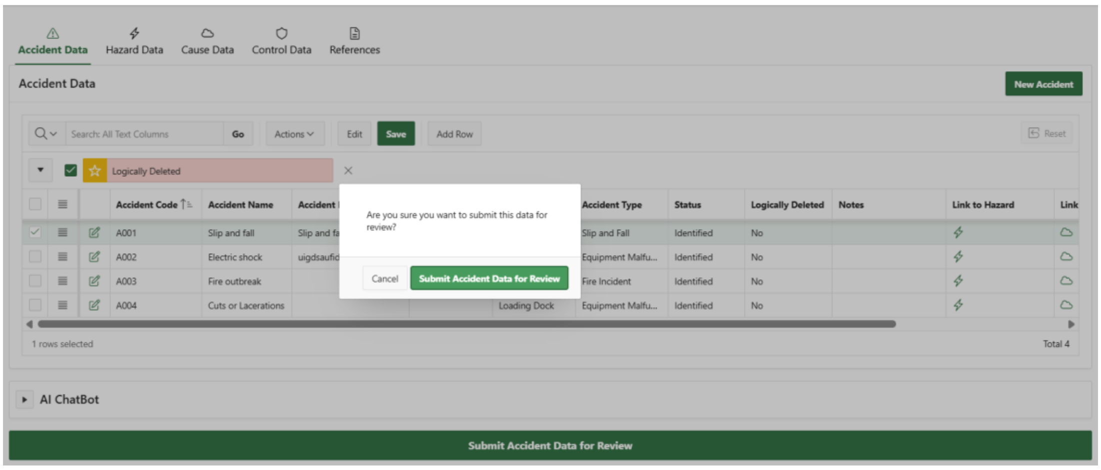

Figure 17 Submit for Review

This updates the status to “Under Review” where the reviewer can see the specific Tab in a reviewer-only Dashboard. When viewing the page, only the reviewer has access to the decision selection shown by Figure 18. If the information is approved, then the status will be updated to “Completed”, otherwise it will change back to “In Progress”.

Figure 18 Review Decision

Following submitting for review or after approval, if the user needs to make changes to the data, they can use the “Make Amendments” button which will change the status back to “In Progress”. The reason for this is due to the projects that this application could be applied to. As the reviewer is taking responsibility for the safety of the product by giving it approval, they need to be aware of all information that they sign off on.

Several Functional requirements have been created to reflect the importance of maintaining the status of entities within the database to ensure data integrity (Figure 19).

```
BEGIN
UPDATE H_PROJECT_TAB
SET STATUS_ID = (SELECT STATUS_ID FROM H_PROJECT_TAB_STATUS WHERE STATUS = 'In Progress')
WHERE TAB_ID = (SELECT TAB_ID FROM H_TAB WHERE TAB_TYPE = 'ACCIDENTS')
AND PROJECT_ID = :P400_PROJECT_ID;

UPDATE H_ACCIDENT_DATA
    SET STATUS_ID = (
        SELECT STATUS_ID
        FROM H_STATUS
        WHERE STATUS_NAME = 'Identified'
    )
    WHERE PROJECT_ID = :P400_PROJECT_ID;

END;
```

Figure 19 PL/SQL updating Tab Statuses

5.4.6. Document Generation

An aspect of the Project identified after performing Gap Analysis (Section 3.3) was the applications need to produce formal sign-off documentation. Figure 20 shows the page used by the user to confirm project information has been captured.

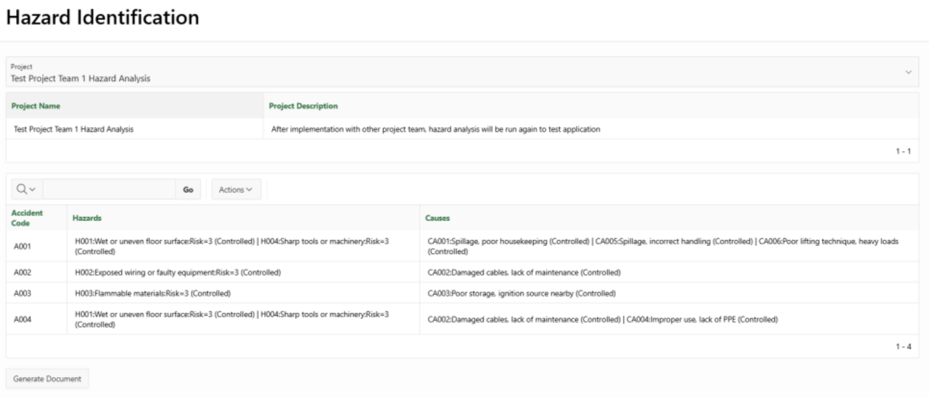

Figure 20 Document Generation Confirmation

Following the user pressing the "Generate Document" button, a file is then downloaded to their device which contains the project information as well as areas for the relevant DAs to sign. The goal of this feature was to reduce the time and effort currently experienced by using the legacy Excel template and having to sanitise check every document that is outputted. To obtain the data used to populate the document, Oracles RESTful service was used. See Section 5.7.2 for a more detailed explanation of how this was achieved.

5.5. Application Components

5.5.1. Plug-ins

Plug-ins are a feature of APEX development and are used to extend the capabilities of Oracle APEX. In this case, plug-ins were considered to provide the document generation function as described in Section 5.4. Plug-ins were provided by united codes APEX Office Print – an architecture diagram showing how the plug-in works is in Figure 21. This option was not used as part of the final project implementation due to the need to extend the project beyond APEX.

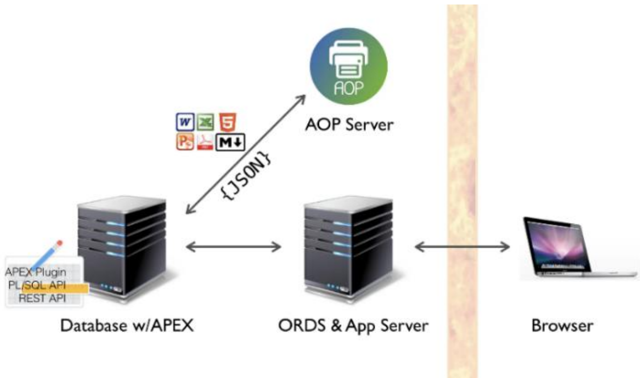

Figure 21 APEX Office Print Architecture - APEX Office Prints (2025).

5.5.2. List of Values

A list of values (LOV) is a static or dynamic definition used to populate page items with values. They can be static, meaning it is based on values the user enters, or dynamic, meaning it is based on a local data source e.g. a table within the database. Due to these pre-defined lists, users can only select from allowed values improving the consistency and accuracy of data. Should the LOVs need to be amended, administration pages available to administrators have been developed to allow for the update of all LOV components.

5.6. Feedback on Design

As previously mentioned in Section 1.1, I have contact with APEX developers who could provide support to me in the form of feedback on the application. This took place after a large amount of development had been completed but before any form of testing was scheduled. Performing the feedback at this stage allowed for further refinement of requirements should there be any feedback regarding features or processes.

As part of the feedback, users were given a log in for the system so that they were able to access the application and view it from both a user and a developer's point of view. The intention for this exercise was to allow full access to the system with no specific requests or guidance, with the user only being able to use their existing knowledge of processes and systems to guide themselves through the application. This was deliberately done to highlight any accessibility features i.e. button placement, text, or common practises that would have otherwise been missed if there were a set of instructions dictating activities (like those from a test case).

Below is the feedback along with a plan on how the suggestions will be implemented into the system along with any amendments to the schedule, if required.

"Overall, your app's looking good. I played around with some of the functionality and a big positive is I didn't experience any errors. That might sound like a fundamental thing, but it's an important thing to get right. Cosmetically, it's looking fine… That type of thing depends on your audience and it can also come down to costs."

This is an indication that the project can be moved into the testing phase following the addressing of any comments. I agree that more could be done cosmetically, going beyond the default cosmetics provided by APEX however, I feel like due to the professional environment that the application could be used in i.e. engineering or defence, it is not necessary at this stage.

5.6.1. Addition of a Landing Page

"Add landing page with info about the app – the current home screen seems to land on a master/detail report and it's not clear what the app's purpose is"

A landing page will allow new users to learn about the app and the information that is able to be captured in it. I believe that this is valuable as if this application is used by people that already know the hazard analysis process, the use of the application is new to them, so it's important to highlight key functionalities to allow for a smooth integration into the system. As a result, the existing Functional Requirement (JAJQ-20 Landing Page) which states that the Project Dashboard Page will be the landing page of the application has bee updated to reflect this new information-based landing page (Figure 22).

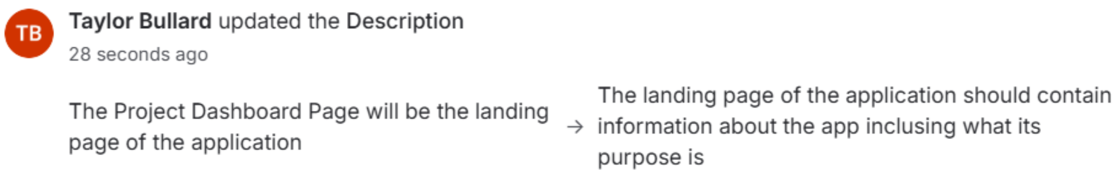

Figure 22 Updating Requirement JAJQ-20 Landing Page

5.6.2. Removing Default APEX columns

"Remove any default APEX columns unless they're needed, e.g. the option to choose a 'Single Row View' isn't really necessary - sometimes less is more"

These are simple to remove and will make the application feel more custom than just using the default APEX columns. Additionally, as highlighted by the comment, the ability to view a single column is not necessary in some reports e.g. the linking pages as it is easier for the user to see all available options before linking target entities. This would not be possible in the single row view default set-up.

5.6.3. Consistent Button Sizing

"Button sizes are stretching the full width of the screen in some cases - my preference is to limit their sizes and keep it consistent through the app"

In keeping with consistency, for the pages that are designed entirely from scratch, and which include these stretched buttons, their size will be limited so that they remain in just one section of the page. As well as it being a preference, if the application was integrated into a "catalogue" or "family" of other applications, both users and developers would expect to see the same appearance and design of features throughout, keeping with familiarity.

5.6.4. Use of Hot Buttons

"Use of 'Hot' buttons… hot buttons should be used when processing data, e.g. submitting a form. For navigation don't use hot buttons and this will distinguish their importance."

This is already a rule of thumb that I was familiar with and tried to make this consistent across the application. I am to review the application to ensure that navigation buttons aren't "hot".

5.6.5. Navigation

"Navigation could be cleaner - I got a little bit lost at times as a new user of the app"

This is an important comment to develop further as it is an existing Non-Functional Requirement (JAJQ-67 Navigation) that the application shall be simple to navigate with a clear navigation menu and page buttons. The ability to navigate the application easily will also be reflected upon following feedback from user testing.

Before then, an effort will be made to map out the different paths that a user can follow to ensure that they can navigate to the pages that they want to from the page they are on. Additionally, the buttons that already exist within the application will be reviewed to ensure that there is a sufficient explanation as to where the button will take the user.

5.7. Additional Functionalities

Sometime into the project, the decision was made to extend the project beyond an Oracle APEX application with additional (more complex) functionalities. Below are the concepts that have been researched as well as their early implementation and how they could possibly but utilised as part of the existing APEX application.

5.7.1. AI Model for Hazard Classification

Within Section 2.4, it was shown AI can be used at different points of the hazard analysis process. This idea comes from the need to identify risk levels for each hazard.

At the early stage of the project, members of a team will come together with ideas for potential hazards, accidents, etc. before later classifying them with a risk level. Having a feature of the application that can accurately and automatically perform this classification saves time in the long-term. Additionally, the feature creates a standardised understanding across all projects in an organisation as similar hazards will be classified the same once a model has been trained.

The following sections will describe the use of lexicons which are to be used in combination with training data to predict a severity and likelihood level for each hazard.

5.7.1.1. Data Collection

As previously mentioned, the idea for this project came from when I was on placement so when reaching out to my placement provider, I asked about the use of their data for the purpose of identifying risk levels. Due to the nature of the data being of a high classification, using their data wasn't possible. Additional research was done to try and gather examples from hazard logs that are available online however these were for projects that had different requirements for risk e.g. infrastructure rather than defence. Those projects that were found did not have enough available and suitable data to put together a sufficiently large enough training data set.

An option that was explored was using Generative Artificial Intelligence (GAI) prompts to create a large enough data set for training. A main drawback from this approach is the quality of data and how the risk level that the LLM assigned to each hazard may not have been accurate, this is something that can be seen later when performing analysis of the classifiers. An attempt was made to sanitise the entries by providing accurate examples to the LLM as well as reviewing lines after they had been generated. The main source of accuracy loss seen in the analysis comes from not having generated sufficiently accurate data at this stage.

5.7.1.2. Use of Semantic Lexicons

A lexicon is a structured knowledge base (like a dictionary) that helps AI models understand and process language by defining keywords and what they mean in the context of what is trying to be achieved. In the case of this project, a list of hazardous keywords i.e. *wet, slippery, hot, failure,* and *error* were each assigned a severity level from 1-5 for the purposes of severity. A list of likelihood related words i.e. *possible, likely, occasional,* and *known* were also assigned a value 1-5 as a likelihood level. This can be used in collaboration with different classifiers to provide a more accurate classification of the hazards. Diagrams showing the effect that these lexicons had on the overall accuracy are to be discussed as part of Section 5.7.1.3.

5.7.1.3. Classifier Analysis

The following classifiers were used analyse their accuracy to help determine the most effective for processing the data and making the hazard predictions:

- MLP Classifier – an artificial neural network made up of an input layer, one or more hidden layers, and an output layer and capable of learning complex and non-linear patterns.

- KNN – A distance-based classifier that works by determining the K closest pieces of training data and selecting the most frequent class among them.
- Linear SVM – A supervised learning algorithm that finds the optimal straight line that best separates classes with the most margin.
- Decision Tree – A tree-structured model that makes decisions by splitting data based on feature values with each node representing a rule and each leaf node representing a prediction.
- Perceptron – A linear classifier which updates weights iteratively based on classification errors to adjust and make corrections.

Each of these were chosen based on my previous experience with classifiers and some language specific models such as Linear SVM, due to working with text. It was found that after refining their attributes, each performed with an accuracy of around 64% as shown by Figure 23.

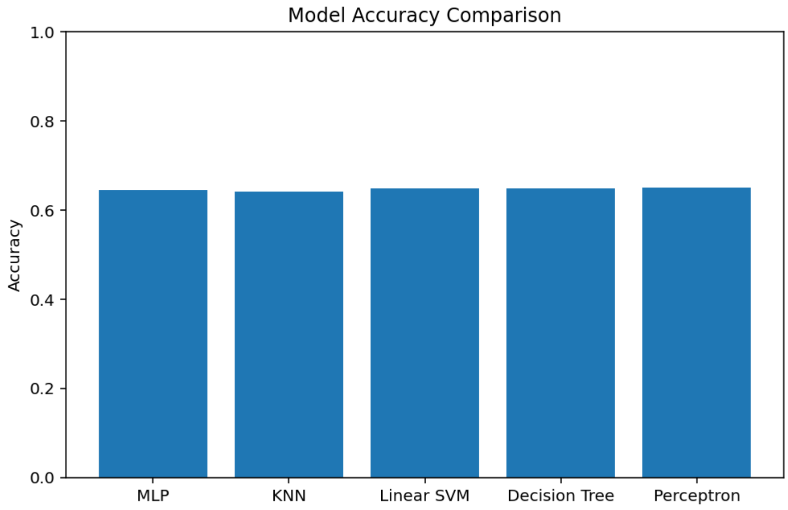

Figure 23 Model Accuracy Comparison

Additionally, when looking at the confusion matrices (Figure 24), Linear SVC, Decision Tree, and Perceptron classifications are occasionally mislabelling the data and predicting a level of 1. This means that despite their accuracy, it is likely that any hazard predicted would output 1.

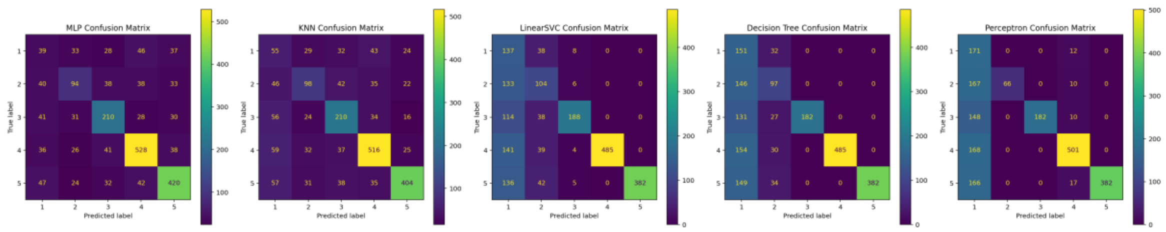

Figure 24 Comparison of Confusion Matrices

As previously mentioned, the use of the lexicon dramatically increases the accuracy of the data. For example, Figure 25 shows that the use of the lexicon produces an accuracy of around 64% whereas with just the classifier, the accuracy is around 20%. This produces an at-chance output, randomly picking a number between 1-5.

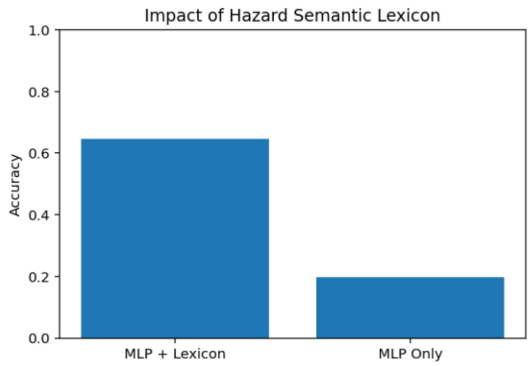

Figure 25 Impact of Lexicon on Accuracy

In Figure 26, these results can be seen further with the confusion matrix without the lexicon (right) showing a random distribution and not a prominent diagonal line like that shown by the MLP with lexicon matrix (left).

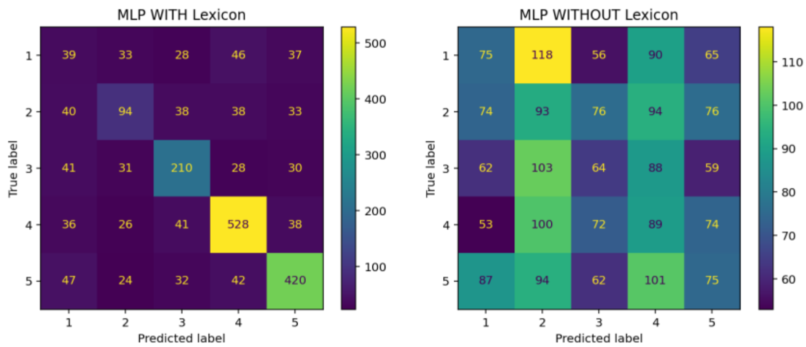

Figure 26 Impact of Lexicon on Accuracy (Confusion Matrices)

5.7.1.3.1. Likelihood Analysis

The same process was followed to predict likelihood however only an accuracy of 21% was achieved (Figure 27). This shows that the lexicon is not suitable and does not contain adequate definition of the likelihood keywords, hence further proving the need for an accurate data set.

```
=== Likelihood Classification Report ===
              precision    recall  f1-score   support

           1      0.241     0.205     0.221       439
           2      0.230     0.174     0.198       391
           3      0.211     0.251     0.229       406
           4      0.224     0.278     0.248       403
           5      0.173     0.166     0.170       361

    accuracy                          0.216      2000
   macro avg      0.216     0.215     0.213      2000
weighted avg      0.217     0.216     0.214      2000

Likelihood Accuracy: 0.216
```

Figure 27 Classification Report for Likelihood

Again, these results are further confirmed by its confusion matrix that appeared randomly distributed (Figure 28). This model will still be used as part of the implementation of the application with the thought of further iterations adjusting the lexicon and training data set.

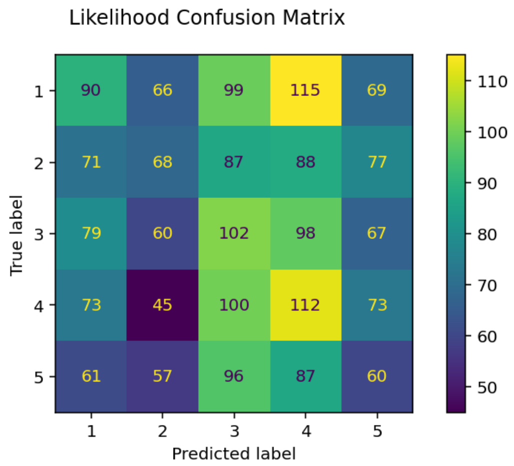

Figure 28 Likelihood Confusion Matrix

Research from Khalilia (2021) expresses the possible reasoning for such low accuracy being lexicons can either include too much wrong information or not enough of the correct information. In the case of this project, including keywords that have incorrect meanings or no relation to the goal of hazard analysis introduces "noise" that can confuse classifiers and reduce their effectiveness. Khalilia (2021) calls this "Overloading". "Underloading" is another issue that can be encountered when not enough correct information is included, creating gaps in the classifier's knowledge. From additional research into the problem, I believe that a future solution that could improve accuracy is by determining likelihood based on the hazard's category, making use of an additional field. This is because the same word e.g. slip can have multiple likelihoods, which is what is causing confusion and a low accuracy.

5.7.1.4. APEX Implementation

The trained models were saved and loaded using Joblib, while Flask was used to host a local server and an HTML user interface for interaction. This interface can then be built using the python code that was originally created via packages and dependencies. Using APEX's page generation and ability to inject html code to display content, an iframe was then embedded in the page to display the localhost webpage (Figure 29). One drawback is the inability to fully test this with external users due to it being hosted locally, restricting the accessibility beyond the host machine. This problem is currently being researched further, and a risk has been captured in the risk register for better visibility.

<iframe src=http://127.0.0.1:5000/ style="width:100%; height:600px; border:none;"> </iframe>

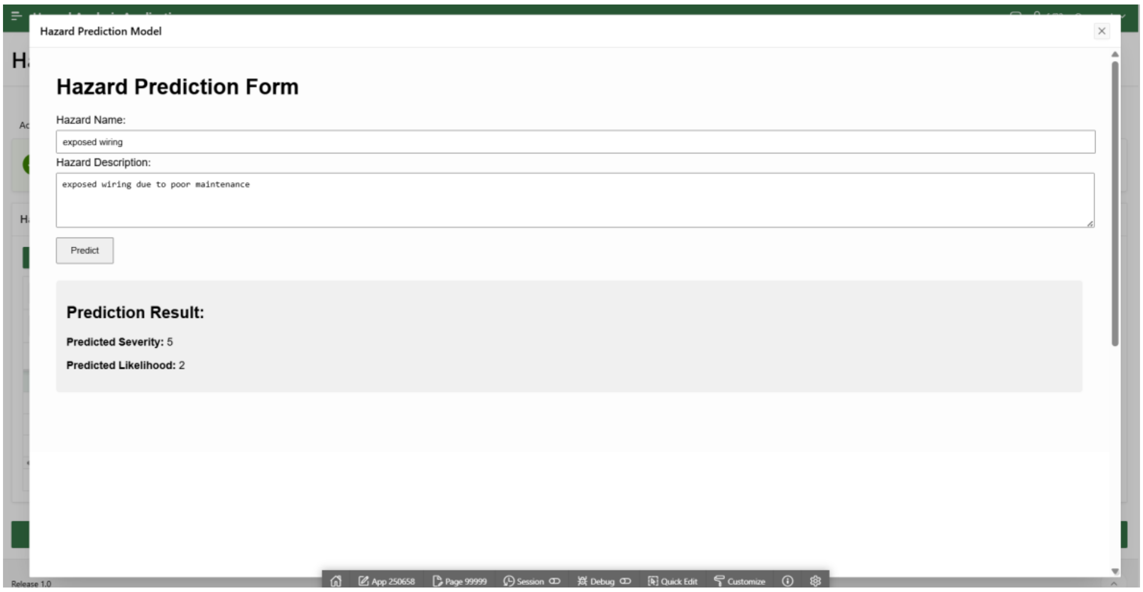

Figure 29 Oracle APEX implementation of AI Model UI

5.7.2. Manual Implementation of Plug-in (document generation)

An implementation of the APEX Office Print plug-in does allow for a document output however it is not fully customisable. The decision was made to perform initial research into alternative ways of how a more customised output could be achieved. Using Oracles RESTful services, modules can be created that will serve SQL queries from the database to a URL. The query stores the data in a. json format so that it can be accessed on a webpage. Following this, a localhost server, created via node.js, hosts a webpage that allows for the viewing of data being passed from the Oracle database. This was then developed further by using JavaScript to generate a .pdf file that displays the information from the previous webpage. The implementation extended the JavaScript file with pdf kit and formatting.

6. Test Plan

The purpose of the test plan is to provide a clear outline for the testing process including the activities that will take place as well as who is responsible for completing them. This is to ensure that all requirements that have previously been defined are associated with a test case so that they can be tested against and deemed compliant.

6.1. Methods

Different approaches of testing were performed to validate the projects' requirements to ensure that they are compliant and meet with pre-defined objectives. The approaches include both Whitebox system testing and standard user (acceptance) testing.

Whitebox system testing consisted of following all test cases to ensure that they could be completed by developers of the system. As part of user (acceptance) testing, initial communication was made with testers to register their information, brief them on the project and test cases (specifically JAJQ-76 Create a Project, JAJQ-78 Data Input and JAJQ-88 Hazard Risk AI Prediction). Testers that followed the test cases used an online form to capture any errors or problems faced during the process so that it could be fixed before Whitebox system testing. The results from the online form were then collated and compiled for analysis. The raw data collected forms part of the supplementary information in the project submission.

6.2. Requirements

To perform the following test cases, Table 3 has been used to capture any additional requirements needed to fulfil the test case and associated requirements.

Test Case	Requirement
All	A device cable of running Google Chrome and Microsoft Edge
JAJQ-92	A stopwatch/timer cable of recording time

Table 3 Testing Requirements

6.3. Test Cases

A full list of all test cases can be found in Appendix F (Test Cases). These are used to inform the step-by-step tasks for the formal testing stage.

6.3.1. User Testing

Feedback gathered from user testing formed evidence for the compliance of Non-Functional Requirements: JAJQ-60 Error Handling, JAJQ-61 Navigation, and JAJQ-62 Design.

7. Test Results and Discussion

7.1. Results and Cross-Validation with Requirements

This section shows the results of the formal testing. A table of all test cases and associated requirements has been complied to provide better visibility as to whether all requirements have been covered by a successful test and can therefore be marked as compliant. This table can be found in Appendix E (Test Case Cross-Validation with Requirements). Evidence of the testing taking place is also detailed in Appendix G (Test Environment).

7.1.1. Example of passed test cases

The purpose of this section is to provide an example of a test case that was produced and completed as part of the Whitebox system testing. For each test case, a spreadsheet was used to compile evidence of the tests and any errors/bugs that were identified (Figure 30).

JAJQ-79 Review Tab	JAJQ-8 Status JAJQ-10 Review JAJQ-11 Link	16/03/2026 10:24	Pass	All steps completed with no errors - inconsistencies with decision box sizes now all conform to one size
JAJQ-83 User Roles	JAJQ-12 User Roles	16/03/2026 10:40	Pass	All steps completed with no errors
JAJQ-88 Hazard Risk AI Prediction	JAJQ-87 AI Prediction	16/03/2026 10:47	Pass	All steps completed with no errors
JAJQ-89 Document Generation	JAJQ-89 Document Generation	16/03/2026 10:51	Pass	All steps completed with no errors - ended up testing with both implementations - localhost and AOP
JAJQ-90 Data Management	JAJQ-9 Data Management	16/03/2026 10:58	Pass	All steps completed with no errors
JAJQ-81 Browser Compatibility	JAJQ-58 Compatibility	16/03/2026 11:00	Pass	All steps completed with no errors
JAJQ-92 Page Load Time	JAJQ-62 Navigation	16/03/2026 11:25	Pass	Completed with an average time of 2.514
JAJQ-82 User Feedback	JAJQ-16 User Feedback	16/03/2026 11:40	Pass	All steps completed with no errors

Figure 30 Test Case Evidence

The test case JAJQ-92 Page Load Time concerns the process described in Section 5.6.5, particularly the time taken to navigate from the Dashboard to a given tab via the redirecting page. The purpose of the test case was to measure this time and ensure that it was reasonable for users. The results were collected in Figure 31 where an average time was calculated. The average was compared to the threshold detailed in the test case description and as the result was within, the test case passed and associated requirement could be marked as compliant.

Collection	Time		Average Time	2.514
1	2.21			
2	2.57			
3	2.24			
4	2.42			
5	2.64			
6	2.66			
7	2.66			
8	2.59			
9	2.34			
10	2.81			

Figure 31 JAJQ-92 Timing Results

7.2. User Testing Results

7.2.1. Introduction

User testing is an extension on the formal testing process which allows users, which are unfamiliar with and have not been involved with the development of the system, to use the application to find bugs. Each tester was provided an account for the system, a list of test cases and a questionnaire, where they made general comments on the completion of the test cases as well as the overall application (all captured as part of Section 7.2.2).

7.2.2. Form Questions

7.2.2.1. Which Web Browser did you use to access the application?

To investigate JAJQ-63 (Browser Compatibility), testers were asked which browser they used so it could be verified that the application works across browsers (specifically Chrome and Edge). There is evidence for the application being tested across both browsers for full coverage.

Figure 32 Browser Question Response

7.2.2.2. Test Case 1: Create a Project - Were you able to successfully complete all of the steps for this Test Case?

All the testers reported that they were able to complete all steps for the test case. The main feedback came in the form of a duplicate project creating itself when a user was creating the first project for a project team. After the bug was first identified, the SQL INSERT query was amended which fixed the bug.

7.2.2.3. Test Case 2: Data Input - Were you able to successfully complete all of the steps for this Test Case?

All the testers reported that they were able to complete all steps for the test case. The main feedback from this test case concerned the navigation. It was highlighted that it would be beneficial to have a “Back to Dashboard” button rather than going through the Navigation Menu each time. This additional button was added during the testing phase to ensure that it was properly tested by users prior to Whitebox testing.

7.2.2.4. **This Test Case only applies to User Testers contacted about testing this functionality** Test Case 3: Hazard Risk AI Prediction - Were you able to successfully complete all of the steps for this Test Case?

From this we saw no problems with all testers reporting that they were able to complete the test case with no errors. In a Microsoft Edge browser, it was noted that an "Apps and Services" pop-up had to be accepted to use the modal window. This was valuable information and the reason as to why the application needed to be tested across different browsers.

7.2.2.5. I found it simple to navigate the application and find the pages required

Additionally, to show the compliance of JAJQ-67(Navigation), testers were asked about their experience navigating the application. In conclusion, we can see from the results below that majority of the users either "Agree" or "Strongly Agree" with majority stating that they found it simple to navigate the application and find the correct pages. The "Disagree" was made along with feedback regarding the navigation which prompted additional development before releasing the application to other user testers. Therefore, we can say that this specific Non-Functional Requirement is compliant.

Figure 33 Navigation Experience Response

7.2.2.6. If you have any other Feedback / comments on the Application then please add it here:

Other feedback mainly included how the test cases were written which led to them continuously being updated with the user testers in mind. For example, specifying where buttons were on the page and the objective of the application as some of the testers were not familiar with Oracle APEX nor the hazard analysis process.

7.2.3. Updates post-testing

As mentioned throughout Section 7.2.2, the purpose of the user acceptance testing was to gather feedback for improvements as well as for functionality. Following the feedback from testers, additional buttons were added after several mentioned that they found difficulty navigating to the Dashboard. These buttons included a direct button on the Landing Page as well as back buttons on each of the Tab Pages.

Further updates were made regarding the fixing of bugs which included updating the condition of the Project INSERT statement to ensure that it was not triggered when the user wanted to just update an existing Project. This specific information was captured as part of a change log where the application repository was kept. This was to ensure that if there were future bugs introduced by changes, the exact version can be rolled back to.

7.3. Non-Compliance

The following requirements were raised as non-compliant during the testing phase meaning the test case failed. This section explains each of these instances as well as their justified removal from the system to ensure complete compliance across all remaining requirements. Additionally, to reflect that the system will not be containing these requirements, their MoSCoW status has been changed to “won’t”.

Requirement	**Justification for non-compliance**
JAJQ-31 Reviewer Decision – Feedback	This feature was originally not included as part of development and has not been raised as an issue with feedback or User Testing. This could be a feature that gets included in future iterations.
JAJQ-53 Data Validation Rules	APEX has rules to ensure data type i.e. no words in a date field or a number field. There were no fields in the application where this needed extending to include the correct validation, but this is something that can be implemented in the future if required.
JAJQ-54 AI Chatbot	Although the feature has been included on the AccidentData page, the focus of the project changed from using APEX and it’s out of the box features to developing more complex and new features. Therefore, it is now not as high a priority and not included as part of formal testing.
JAJQ-70 Change Password in Application	Not necessary at a proof-of-concept stage. However, will be necessary to implement into the application should it go into service and is regularly used by a userbase.

JAJQ-75 User Guide	Not necessary at this level however can be considered should the application go into service as there will need to be some supplementary training due to it being a new tool.

Table 4 Justification for Non-Compliance

Therefore, following the formal testing stage, apart from justified instances of non-compliance, the results show compliance of all requirements that are essential. These are to be included as part of the initial deployment of the application.

8. Discussion

This project has provided an alternative to the legacy Excel template used by my placement organisation for hazard analysis. Following suggestions by Kvet (2024) in support of Oracle APEX as a development tool, I have created a more bespoke application that fulfils the organisations' needs and showcased the effectiveness of low-code application tools.

8.1. Evaluation of objectives

The main aim for this project was to develop an application that is suitable for use within an organisation to complete hazard analysis on any project. Following user testing and the feedback obtained from that, I can state that it is usable and accessible as well as functional against identified requirements. Additionally, the system meets requirements that were obtained from assessing organisation needs.

Moreover, it provides a greater, complex functionality that furthers the area of hazard analysis via the implementation of AI through hazard classification. Overall, this project has been successful upon reflection of the objectives stated at the start of the project. This has been shown by Table 5 which includes justifications for how each of these objectives have been met.

Objective	**Has objective been met?**	**Additional Comments/Justification**
To perform academic research into the chosen subject area and surrounding	Met	Full Literature Review performed as formative task to compile project

tools/processes concerning Hazard Analysis and Application Development		themes with important themes used in final project portfolio
To create a set of Functional and Non-Functional Requirements that are to be used to inform the development of the system	Met	Jira used to compile a full set of parent-child related Functional and Non-Functional Requirements. Non-Compliant requirements were addressed.
To produce a comprehensive set of technical documentation in line with my placement organisation. This includes Project Management Plan, System Requirements Specification, System Design Document, Test Plan and Test Results.	Met	This document is based on the structure outlined in the objective whilst also working to the assessment brief as part of the Final Year Project
To produce a functioning Oracle APEX application (along with a suitable database structure) which fits the purpose of Hazard Analysis	Met	An application has been made using Oracle APEX (along with supporting database). The design has been reviewed by developers and feedback given
To develop sufficiently complex features that can be incorporated within an Oracle APEX Application and that meet the project requirements for CMP accreditation	Partially Met	See Section 5.7.1 and 5.7.2. More information below about what more could be done
To perform thorough and robust "formal testing" which includes a combination of both Developer and User Testing	Met	User Testing was used to inform design and identify bugs with Formal Testing confirming Requirement compliance
To seek guidance and feedback from developers that use Oracle APEX from my previous placement to ensure the system is well built and to an Industry Standard	Met	Feedback provided from Oracle APEX developer that I worked with during my placement. A full list of feedback and actions were captured

Table 5 Evaluation of Objectives

An objective that I believe could have been worked on further was regarding "sufficiently complex features that can be incorporated with an Oracle APEX Application". Despite implementing two features, the justification for marking this as "Partially Met" is that I could have found other complexities rather than recreating an already existent APEX plug-in. In terms of the theme of the project, there was not an additional functionality that came to mind regarding a complex feature. I feel like this is a potential drawback of Oracle APEX. Due to its nature of being a low-code application primarily focusing on data management, there isn't as much flexibility with integrating other features as opposed to a fully custom-coded application.

This is a sentiment expressed by Gorumutchu (2023), who highlights APEX's limited integration options and ability to interact with other systems. If this project were to be produced again, greater research could initially be performed to evaluate the different development platforms (whether that be low-code or fully custom-code). Evaluating the advantages and disadvantages would allow for a more informed decision on the development platform used.

8.2. Future Work

On the other hand, apart from improving upon project objectives, there are other aspects of the system that can be enhanced or developed in the future. As highlighted in Section 7.3, non-compliant features can be used as a basis for any future enhancements. Some of these features (such as JAJQ-70 Change Password in Application and JAJQ-75 User Guide) are necessary or would benefit the deployment of the application if it were used on a project by customers.

As previously identified in Section 5.7.1, there would be a need for improvement of the model training data as well as the supporting lexicons. Should this application and feature be used in industry, there would be an existing database of previous projects that can be used as the training data. Following this, the lexicon, which has shown to dramatically improve accuracy, can be better suited for the language of the organisation and the new training data. Although I have shown an early demonstration of how AI can be utilised in the hazard analysis process by aiding with the classification of hazard risk, it is important to consider concerns over data reliability (Diemert and Weber, 2023; Qi, Zhao, Khastgir and Huang, 2025). Data generated through GAI for the purpose of training the model and lexicon introduced the limiting factors of "overfitting" and "underfitting" which was seen in the lower accuracy rates in model analysis (Section 5.7.1.3). Qi, Zhao, Khastgir and Huang (2025) expressed that the use of language models would only produce reliable results when combined with the knowledge of an SME. Therefore, with the AI implementation of hazard classification, especially at this stage of its development, it would be better suited as a tool to suggest probability and severity levels.

It can then be consulted by an SME, forming a kind of “Co-Hazard Analysis” as previously proposed by Diemert and Weber (2023).

Additionally, if this project were to be adopted by a company in the Engineering/Defence sector, there is compliance that must be followed both internally and by the MOD. As previously mentioned, MOD (2020) notes that Excel templates are suitable (Section 2.2) for performing hazard analysis however if an application is used then it must be applied in a way that will be most effective in supporting the safety risk management process. At this current deployment, the application would not be suitable for use on real-world projects and additional development would need to be done to ensure that both the internal and external compliance requirements are met as well as security. If this application is considered by an organisation, then this additional requirement gathering would be performed as part of a next iteration.

Furthermore, with regards to the development and implementation of the additional features, the document generation and risk prediction functionalities are currently executed via localhost. This means that if the application were to be deployed to a large user base, further development will be required to make the application more feasible. In such a scenario, hosting these services on a centralised web server would be more appropriate than requiring each individual user to run a local instance on their device.

8.3. Conclusion

To conclude, from the project’s background and motivation to methodology and implementation, I have demonstrated a solution to replace an inefficient legacy Excel template approach used by my placement organisation. I have shown the abilities of low-code application tools via Oracle APEX (Kvet, 2024) whilst furthering the field of hazard analysis by introducing AI and demonstrating how it can be used to classify risk through the use of lexicons and model training (Diemert and Weber, 2023; Qi, Zhao, Khastgir and Huang, 2025). By measure of objectives, this project has been a success due to identifying, managing and confirming requirements through a robust and thorough testing phase alongside feedback provided from developers working in industry. Future work should address non-compliance and inefficiencies highlighted, such as the AI classification model training data reliability and general security practices, to prepare it for use in a professional organisation.

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Copyright Acknowledgement - In this document, material from a UK Ministry of Defence guide has been used. It is reproduced under the terms of the UK's Open Government Licence https://www.nationalarchives.gov.uk/doc/open-government-licence/version/3/

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10. Appendices

10.1. Appendix A (Functional Requirements)

Requirement ID	Requirement Name	Description	Tag	Version	MoSCoW
JAJQ-1	Manage Projects	Users will be able to create and edit Projects	Projects	Functional_Requirements_V1	Must
JAJQ-17	Project Dashboard	All projects will be displayed on a Project Dashboard	Projects	Functional_Requirements_V1	Must
JAJQ-18	Project User Groups	Only projects assigned to a user's user group will be visible to them on the Dashboard	User Groups	Functional_Requirements_V1	Must
JAJQ-19	Delete Projects	Projects can be set to logically deleted	Projects	Functional_Requirements_V1	Could
JAJQ-20	Landing Page	The landing page of the application should contain information about the app including what its purpose is	Dashboard	Functional_Requirements_V1	Could
JAJQ-21	Tab Generation	Creating a new project will create all tabs necessary for Hazard Analysis to then be filled out by the User	Tabs	Functional_Requirements_V1	Must
JAJQ-22	Add Data	Users will be able to add information to the Data Tabs (Hazards, Accidents, Causes, Controls and References)	Tabs	Functional_Requirements_V1	Must
JAJQ-23	Data Interactive Grid	An Interactive Grid will be used to display and edit Data Hazards, Accidents, Causes, Controls and References) for a selected project	Interactive Grids	Functional_Requirements_V1	Must
JAJQ-24	Interactive Grids	Interactive Grids will allow for Users to create custom reports depending on the data they want to see	Interactive Grids	Functional_Requirements_V1	Must
JAJQ-25	Data Status	Data Tabs must have the Status of either: In Progress, In Review, Completed	Status	Functional_Requirements_V1	Must
JAJQ-26	Data Logically deleted	Data Entries (Hazards, Accidents, Causes, Controls and References) can be set to logically deleted	Data Management	Functional_Requirements_V1	Could

JAJQ-27	Data to Review	Data Tabs can be set to Review	Review	Functional_Requirements_V1	Must
JAJQ-28	Data in Review	Data Tabs that are in Review cannot be edited by a User	Review	Functional_Requirements_V1	Must
JAJQ-29	Editing Data in Review	Data Tabs that are in Review must be set to In Review to be edited by a User	Review	Functional_Requirements_V1	Must
JAJQ-30	Reviewer Decision - Data	A reviewer can Approve or Reject a Data Tab	Review	Functional_Requirements_V1	Must
JAJQ-31	Reviewer Decision - Feedback	A reviewer could provide feedback to accompany a review decision	Review	Functional_Requirements_V1	Won't
JAJQ-32	Rejected Data Tab	Rejected Data Tabs will have their status set to In Progress	Status	Functional_Requirements_V1	Must
JAJQ-33	Approved Data Tab	Approved Data Tabs will have their status set to Comeplete	Status	Functional_Requirements_V1	Must
JAJQ-34	Linking Data	Data from different Tabs can be linked together i.e. AccidentHazard	Link	Functional_Requirements_V1	Must
JAJQ-68	Linking Data – Modal Page	Data can be linked to other day via a checkbox modal page	Link	Functional_Requirements_V1	Could
JAJQ-35	Link Status	Link Tabs must have the Status of either: In Progress, In Review, Completed	Status	Functional_Requirements_V1	Must
JAJQ-36	Link to Review	Link Tabs can be set to Review	Review	Functional_Requirements_V1	Must
JAJQ-37	Reviewer Dashboard	Reviewers will have a dashboard showing pending review items	Dashboard	Functional_Requirements_V1	Could
JAJQ-38	Link in Review	Link Tabs that are in Review cannot be edited by a User	Review	Functional_Requirements_V1	Must
JAJQ-39	Editing Links in Review	Link Tabs that are in Review must be set to In Review to be edited by a User	Review	Functional_Requirements_V1	Must
JAJQ-40	Reviewer Decision - Link	A reviewer can Approve or Reject a Link Tab	Review	Functional_Requirements_V1	Must
JAJQ-41	Rejected Link Tab	Rejected Link Tabs will have their status set to In Progress	Status	Functional_Requirements_V1	Must

JAJQ-42	Approved Link Tab	Approved Link Tabs will have their status set to Complete	Status	Functional_Requirements_V1	Must
JAJQ-43	User Roles	The Application will have the roles: User, Reviewer and Administrator	User Roles	Functional_Requirements_V1	Must
JAJQ-44	Manage User Roles	Admins can manage users assigned to User Roles	User Roles	Functional_Requirements_V1	Must
JAJQ-45	User Groups	Users can be assigned to a User Group	User Group	Functional_Requirements_V1	Must
JAJQ-46	Manage User Groups	Admins can add and remove Users from a User Group	User Group	Functional_Requirements_V1	Must
JAJQ-47	Project User Groups	User Groups will be assigned to a project	User Group	Functional_Requirements_V1	Must
JAJQ-48	Manage LOVs	Values displayed in LOVs will be managed by Administrators	Data Management	Functional_Requirements_V1	Could
JAJQ-49	User Login	Users will be able to login to the application via a login page	Login	Functional_Requirements_V1	Must
JAJQ-50	Document Generation Query	Query of all inputted data to populate a document	Document Generation	Functional_Requirements_V1	Must
JAJQ-51	Document Generation Output	A .docx file is generated for a given project	Document Generation	Functional_Requirements_V1	Could
JAJQ-52	Download generated document	The generated documents will be downloaded and made available to the user	Document Generation	Functional_Requirements_V1	Could
JAJQ-53	Data Validation Rules	Fields will have validation rules (e.g. required, format checks)	Data Management	Functional_Requirements_V1	Won’t
JAJQ-54	AI Chatbot	Users could use an AI chatbot to access information about a project to assist in writing Hazard Analysis	AI Chatbot	Functional_Requirements_V1	Won’t
JAJQ-55	User Feedback	Users of the application will be able to provide feedback on the application	User Feedback	Functional_Requirements_V1	Could
JAJQ-56	View User Feedback	Administrators will be able to view Feedback given from Users	User Feedback	Functional_Requirements_V1	Could

JAJQ-57	Manage User Feedback	Administrators will be able to manage User Feedback by updating its status and addressing comments	User Feedback	Functional_Requirements_V1	Could
JAJQ-70	Change Password in application	Users will be able to change their password once logged into the application by providing a current and an updated password	Login	Functional_Requirements_V1	Won't
JAJQ-84	AI Hazard Prediction	If a User gives a Hazard Name and Hazard Description, a prediction for Hazard Severity and Likelihood will be made	AI Prediction	Functional_Requirements_V1	Could
JAJQ-85	AI Modal Page	The Hazard Prediction functionality will be shown as a pop-up modal page	AI Prediction	Functional_Requirements_V1	Could

10.2. Appendix B (Non-Functional Requirements)

Requirement ID	Requirement Name	Description	Tag	Version	MoSCoW
JAJQ-63	Browser Compatibility	The application shall be compatible with the latest versions of Chrome and Edge.	Compatibility	Non-Functional_Requirements_ V1	Must
JAJQ-64	Supporting Documentation	A set of supporting documentation will be written to provide support for future development and releases as well as providing in service support	Documentation	Non-Functional_Requirements_ V1	Must
JAJQ-65	Error Handling	The system shall display user-friendly error messages and log technical details for developers.	Error Handling	Non-Functional_Requirements_ V1	Could
JAJQ-66	Application Design	The application design shall be accessible whilst using APEXs existing design tools.	Design	Non-Functional_Requirements_ V1	Must
JAJQ-67	Navigation	The application shall be simple to navigate with a clear navigation menu and page buttons	Navigation	Non-Functional_Requirements_ V1	Must
JAJQ-74	Version History	The version number of the application will be visible to users	Version	Non-Functional_Requirements_ V1	Could

JAJQ-75	User Guide	A supporting documentation will be available to users which demonstrates the main functionalities of the application	User Guide	Non-Functional_Requirements_ V1	Won’t
JAJQ-91	Navigation Time	Time for a tab page to load from the Dashboard should be less than 5 seconds.	Navigation	Non-Functional_Requirements_ V1	Should

10.3. Appendix C (ER Diagram)

For a complete definition of all tables see Appendix H (Table Definition). Due to the size of the entire system, the diagram has been separated in sections (see captions) to best present the tables relations. Definition of the tables is in the supplementary project documentation.

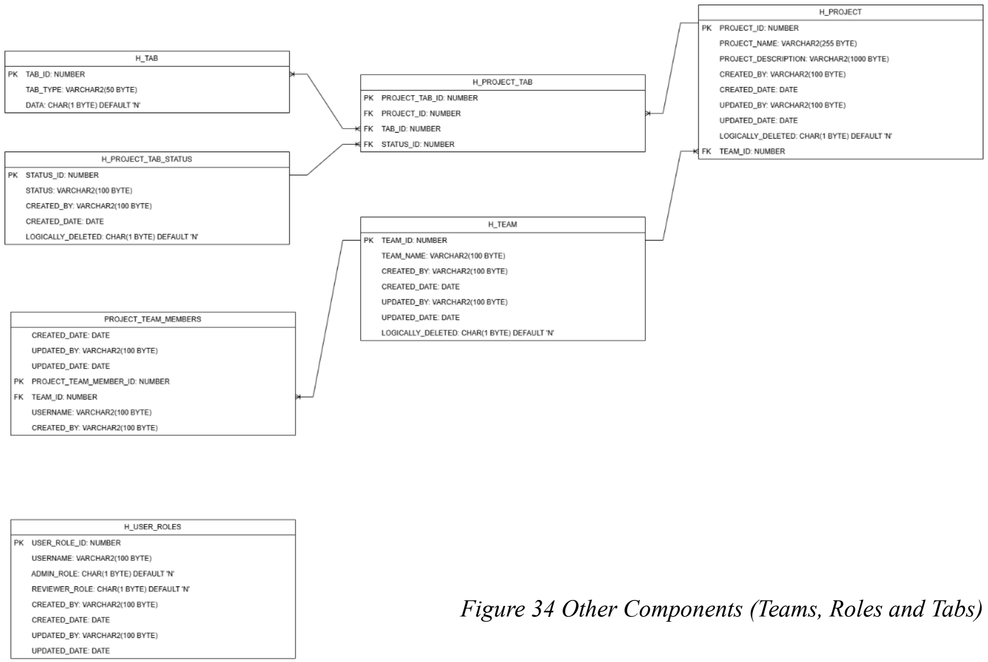

Figure 34 Other Components (Teams, Roles and Tabs)

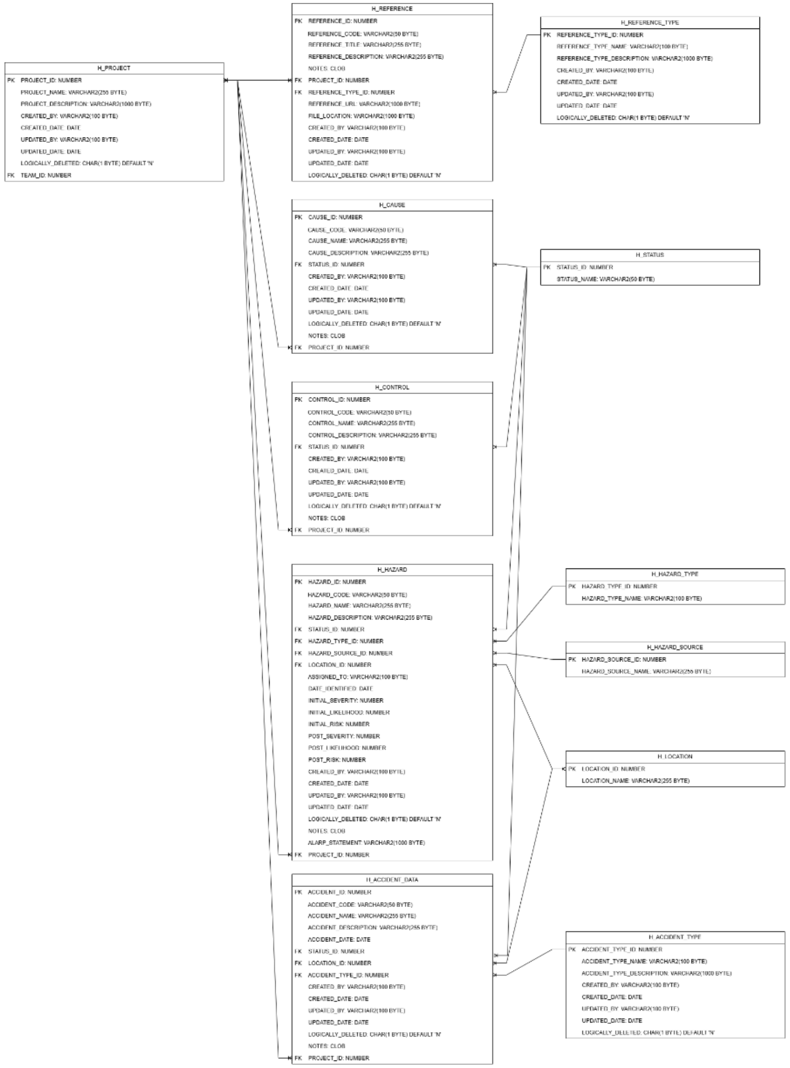

Figure 36 Data

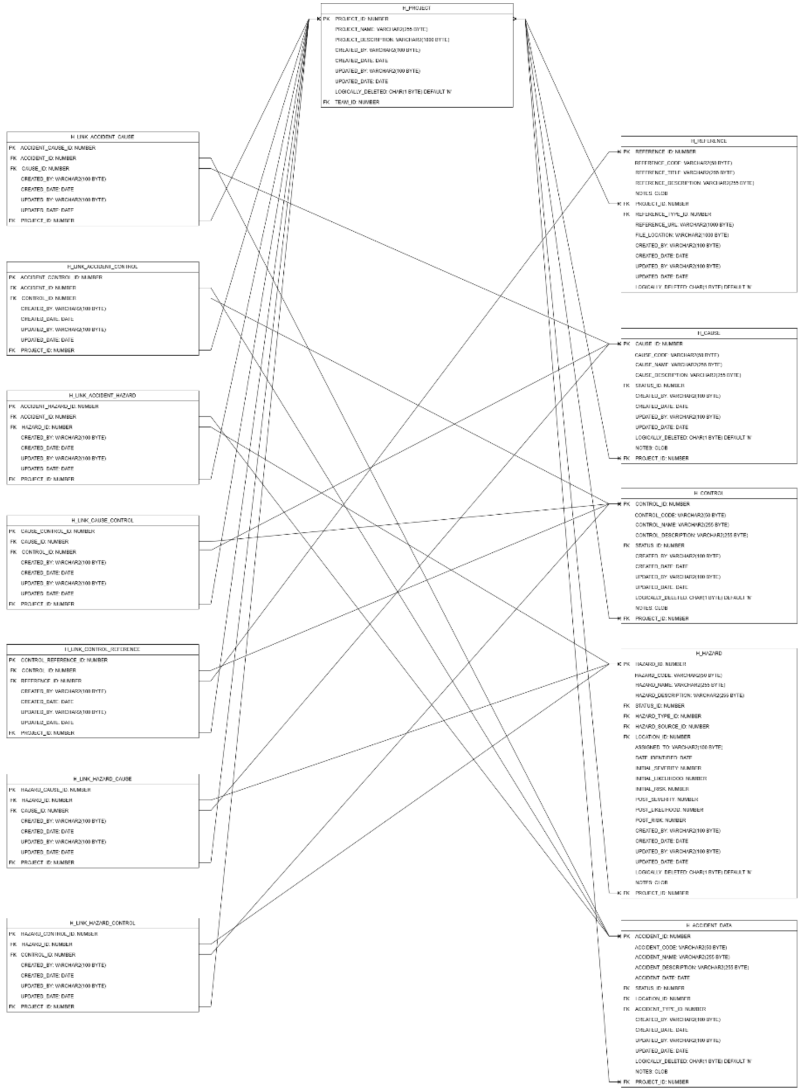

Figure 35 Linking Data

10.4. Appendix D (Gap Analysis)

Area of Focus	Current State	eCassandra	ISCaDE™ Pro	Proposed State (Oracle APEX)	Gap	Action Plan
Capturing of Data	Data must be manually inserted into tabs in an Excel spreadsheet	Pages for users to interface the creation of new data	Spreadsheet style data input and representation	Validation can be implemented as well as existing regions i.e. interactive grids for data management	Need for a tool with greater data interface for editing and linking data	Transition to tool-based solution that has structured input
Linking Data	On the relevant tabs, a dropdown of all related data tabs is available to be chosen from – there's no input validation preventing duplicates	Built-in relational linking with validation	Strong relational database linking	Validated relational database tables linking to front end pages for easy user interfacing	Risk of duplicate or incorrect links	Implement validation rules
Generating required documentation for sign off	The excel template was created with embedded formulas that require sanitation after generating documents to remove unnecessary items added in the query	Does not have a built-in feature to automatically generate formal documentation from its database	Ability to publish the documentation and audit trail of the safety analysis process	Plug-ins provide additional functionality to support application development and include new features	A system them generates Hazard documentation with less clean up than existing solution	Research documents generation plug-ins to provide the functionality
Knowledge of solution	Excel spreadsheets are commonly used in organisations however the maintenance of queries and embedded formulas may only be known by a select few people	Is only known to SMEs that work with project safety as a large part of their work	Largely unknown as other tools (such as eCassandra) are a preferred tool	Similar organisation tools used and developers already in place with knowledge on how to support an application	Largely untrained group of people that perform hazard analysis so using a new tool would lead to better adoption.	Adopt a new tool rather than training to understand Excel

MOD opinion on application for Hazard Analysis	Is possible to maintain Hazard Analysis for smaller scale projects	MOD preferred application for Hazard Analysis	Can be used for MOD related projects	As there are existing applications, those would be preferred over a single organisation using a different application	Although not necessary for non-defence projects, having a system that is adequate to deal with the project complexity would be desirable	Develop an application in line with MOD requirements and functionalities of eCassandra and ISCaDE™ Pro
Sending data for review	A copy of the spreadsheet or link to its file location will have to be sent to the reviewer. Alternatively, a meeting between the originator and reviewer of the document may be held to have more of a discussion-based review	Ability to regularly review in line with MOD guidance	Visibility of information, processes and tools allowing review of the safety case by all stake holders	Tabs can be sent for review to be assessed by a reviewer	A functionality to allow for information to be sent for review and reviewed by a reviewer within the same application and not having to use external tools to send messages/information	Reviewer specific dashboard and pages with buttons and actions only visible to users with a user role
Version Control of Data	Different versions of the spreadsheet may be created over time and rely on the organisations policy on document updates i.e. using OneDrive to check-out and check-in documents to prevent work being done simultaneously	Data contained with a database meaning all data being uploaded and amended is version controlled	Data contained with a database meaning all data being uploaded and amended is version controlled	Tables in Oracle APEX can contain auditing columns such as CREATED_BY and UPDATED_BY to update via triggers on changes to data	Limited visibility in Excel	Migrate to system with auditing capabilities and enforce version control through read only fields
Tracking changes	Version History can be seen for an Excel spreadsheet if it was upload to a service like OneDrive	Logs inserts, updates, and deletes to data in a log format similar to a commit log	Form states: Used in the background, these forms keep track of user actions	Tables in Oracle APEX can contain auditing columns such as CREATED_BY and UPDATED_BY to update via triggers on changes to data	Limited visibility in OneDrive	Ensure system has an audit trail with dedicated pages

10.5. Appendix E (Test Case Cross-Validation with Requirements)

Test Case	Requirement Parent	Requirement	Result	Comment
JAJQ-76 Create a Project	JAJQ-3 Projects	JAJQ-1 Manage Projects	Pass	Verified by test case
		JAJQ-17 Project Dashboard	Pass	Verified by test case
		JAJQ-19 Delete Projects	Pass	Verified by test case
	JAJQ-4 User Groups	JAJQ-18 Project User Groups	Pass	Verified by test case
		JAJQ-45 User Groups	Pass	Verified by test case
		JAJQ-46 Manage User Groups	Pass	Verified by test case
		JAJQ-47 Project User Groups	Pass	Verified by test case
	JAJQ-5 Dashboard	JAJQ-20 Landing Page	Pass	Verified by test case
		JAJQ-37 Reviewer Dashboard	Pass	Verified by test case
	JAJQ-6 Tabs	JAJQ-20 Tab Generation	Pass	Verified by test case
		JAJQ-21 Add Data	Pass	Verified by test case
	JAJQ-13 Login	JAJQ-49 User Login	Pass	Verified by test case
		JAJQ-70 Change Password in Application	Fail	Non-Compliance as not included as part of development – out of scope for proof of concept
JAJQ-77 User Groups	JAJQ-4 User Groups	JAJQ-18 Project User Groups	Pass	Verified by test case
		JAJQ-45 User Groups	Pass	Verified by test case
		JAJQ-46 Manage User Groups	Pass	Verified by test case
		JAJQ-47 Project User Groups	Pass	Verified by test case
JAJQ-78 Data Input	JAJQ-6 Tabs	JAJQ-20 Tab Generation	Pass	Verified by test case
		JAJQ-21 Add Data	Pass	Verified by test case
	JAJQ-7 Interactive Grids	JAJQ-23 Data Interactive Grid	Pass	Verified by test case
		JAJQ-24 Interactive Grids	Pass	Verified by test case
	JAJQ-9 Data Management	JAJQ-26 Data Logically Deleted	Pass	Verified by test case
		JAJQ-48 Manage LOVs	Pass	Verified by test case

		JAJQ-53 Data Validation Rules	Fail	Non-Compliance as feature already included by APEX
	JAJQ-10 Review	JAJQ-27 Data to Review	Pass	Verified by test case
		JAJQ-28 Data in Review	Pass	Verified by test case
		JAJQ-29 Editing Data in Review	Pass	Verified by test case
		JAJQ-30 Reviewer Decision - Data	Pass	Verified by test case
		JAJQ-31 Review Decision - Feedback	Fail	Non-Compliance as not included as part of development
		JAJQ-36 Link to Review	Pass	Verified by test case
		JAJQ-38 Link in Review	Pass	Verified by test case
		JAJQ-39 Editing Links in Review	Pass	Verified by test case
		JAJQ-40 Reviewer Decision - Link	Pass	Verified by test case
	JAJQ-11 Link	JAJQ-34 Linking Data	Pass	Verified by test case
		JAJQ-68 Linking Data – Modal Page	Pass	Verified by test case
JAJQ-79 Review Tab	JAJQ-8 Status	JAJQ-25 Data Status	Pass	Verified by test case
		JAJQ-32 Rejected Data Tabs	Pass	Verified by test case
		JAJQ-33 Approved Data Tabs	Pass	Verified by test case
		JAJQ-35 Link Status	Pass	Verified by test case
		JAJQ-41 Rejected Link Tab	Pass	Verified by test case
		JAJQ-42 Approved Link Tab	Pass	Verified by test case
	JAJQ-10 Review	JAJQ-27 Data to Review	Pass	Verified by test case
		JAJQ-28 Data in Review	Pass	Verified by test case
		JAJQ-29 Editing Data in Review	Pass	Verified by test case
		JAJQ-30 Reviewer Decision - Data	Pass	Verified by test case

		JAJQ-31 Review Decision - Feedback	Fail	“”
		JAJQ-36 Link to Review	Pass	Verified by test case
		JAJQ-38 Link in Review	Pass	Verified by test case
		JAJQ-39 Editing Links in Review	Pass	Verified by test case
		JAJQ-40 Reviewer Decision - Link	Pass	Verified by test case
	JAJQ-11 Link	JAJQ-34 Linking Data	Pass	Verified by test case
		JAJQ-68 Linking Data – Modal Page	Pass	Verified by test case
JAJQ-81 Browser Compatibility	JAJQ-58 Compatibility	JAJQ-63 Browser Compatibility	Pass	Verified by test case
JAJQ-82 User Feedback	JAJQ-16 User Feedback	JAJQ-55 User Feedback	Pass	Verified by test case
		JAJQ-56 View User Feedback	Pass	Verified by test case
		JAJQ-57 Manage User Feedback	Pass	Verified by test case
JAJQ-83 User Roles	JAJQ-12 User Roles	JAJQ-43 User Roles	Pass	Verified by test case
		JAJQ-44 Manage User Roles	Pass	Verified by test case
JAJQ-88 Hazard Risk AI Prediction	JAJQ-87 AI Prediction	JAJQ-84 Hazard AI Prediction	Pass	Verified by test case
		JAJQ-85 AI Modal Page	Pass	Verified by test case
JAJQ-89 Document Generation	JAJQ-14 Document Generation	JAJQ-50 Document Generation Query	Pass	Verified by test case
		JAJQ-51 Document Generation Output	Pass	Verified by test case
		JAJQ-52 Download Generated Document	Pass	Verified by test case
JAJQ-90 Data Management	JAJQ-9 Data Management	JAJQ-26 Data Logically Deleted	Pass	Verified by test case
		JAJQ-48 Manage LOVs	Pass	Verified by test case

		JAJQ-53 Data Validation Rules	Fail	“”
JAJQ-92 Page Load Time	JAJQ-62 Navigation	JAJQ-67 Navigation	Pass	Verified by test case
		JAJQ-91 Navigation Time	Pass	Verified by test case

10.6. Appendix F (Test Cases)

10.6.1. Create a Project

JAJQ-76 Create a Project			
Step	Task	Expected Outcome	Success
1	Tester will login with the account information provided for them	Tester will be taken to the Application information Landing page	
2	Tester will navigate to the Project Dashboard and press New Project and fill in the fields on the page	Tester will be taken to information page and then navigate to the Project Dashboard page and fill in fields	
3	Tester will press Create Project button	Tester will have no errors and be taken back to the Project Dashboard page	
4	Tester will press the inspect (magnifying glass) icon for the associated project under project name	Tester will be taken to project page where all information will be in place	
5	Tester will update a field and apply changes	Tester will have no errors and be taken back to the Project Dashboard page	
6	Tester will return to project page and press logically delete	Tester will be taken back to the Project Dashboard page where a visual identifier that the project has been logically deleted will be seen	

10.6.2. User Groups

JAJQ-77 User Groups			
Step	Task	Expected Outcome	Success

1	Tester will login with the account information provided for them	Tester will be taken to the Application information Landing page	
2	Tester will navigate to the Administration Tab > User Teams	Tester will be able to see these tabs and navigate to the page	
3	Tester will press Create	Modal Page with Team Name will appear	
4	Tester will type in a Team Name and then submit the page	No errors and return to Team Page	
5	Tester will navigate to Manage Project Teams	Tester will be able to see these tabs and navigate to the page	
6	Tester will then assign a User to a Team	Tester will have used the Interactive Grid drop down to assign a User to a Project Team	

10.6.3. Data Input

JAJQ-78 Data Input			
Step	Task	Expected Outcome	Success
1	Tester will login with the account information provided for them	Tester will be taken to the Application information Landing page	
2	Tester will navigate to the Project Dashboard Page and then to a Project on the Dashboard by using the select buttons (circles) on the far left of the project area	Tester will be able to see the project and its associated tabs	
3	Tester will press on the HAZARD tab that is In Progress	Tester will be shown a loading screen and then the tab of their choosing	
4	Use the Interactive Grid to fill out a line of the tab information and press save	Interactive Grid will save with no errors	
5	Edit the same line and press save	Interactive Grid will save with no errors	
6	Use the actions tab > columns to add or remove columns	Columns will be added/removed	

7	Following the addition of the data, the Tester will press the “Submit Data For Review Button”	A button will appear for make amendments and the submit button will be removed	
8	Tester will repeat steps 4-7 again in the ACCIDENTS tab using the tab navigation at the top of the page to do so	Tester will be navigated to another tab information page	
9	Using the Link to Hazard icon, check some of the boxes (linking the tab data together)	All project data for the selected tab will be shown and be available to select	
10	Use the Go to “Go to AccidentHazard” button on the modal page to check that the same data is there on the link page	Same data to be shown	
11	Following the confirmation of the data, the Tester will press the "Submit For Review" Button	A button will appear for make amendments and the submit button will be removed	

10.6.4. Review Tab

JAJQ-79 Review Tab			
Step	Task	Expected Outcome	Success
1	Tester will login with the account information provided for them	Tester will be taken to the Application information Landing page	
2	Tester will navigate to a Review Dashboard	Tester will be taken to the Reviewer Dashboard	
3	Press the navigate icon on the tab of the project for Review	Tester will be taken to tab page	
4	Select Reject from the drop-down menu and Submit Review	Tester will remain on the page	
5	Press submit for Review button	The page will reload	
6	This time, select Approve from the drop-down menu and Submit Review	Page will reload with a Review Complete message	
7	Repeat steps 2-6 for a Linking tab	No errors, expected outcome to remain the same	

10.6.5. User Roles

JAJQ-83 User Roles			
Step	Task	Expected Outcome	Success
1	Tester will login with the account information provided for them	Tester will be taken to the Application information Landing page	
2	Tester will navigate to the Administration Tab > User Roles	Tester will be able to see these tabs and navigate to the page	
3	Tester will amend their Role by changing Reviewer to Yes and press save	Confirmation that change has been saved	
4	Tester will navigate to Project Dashboard and should see Reviewer Dashboard in the navigation bar	Reviewer Dashboard tab visible in navigation bar	
5	Navigate back to the Administration Tab > User Roles	Tester will be able to see these tabs and navigate to the page	
6	Tester will amend their Role by changing Reviewer to No and press save	Confirmation that change has been saved	
7	Tester will navigate to Project Dashboard and should not be able to see Reviewer Dashboard in the navigation bar	Reviewer Dashboard tab hidden in navigation bar	

10.6.6. Hazard Risk AI Prediction

Prior to this test, the tester must ensure the device they are usings contains the localhost implement of the AI Prediction code i.e. is given the code or using developer's device.

JAJQ-88 Hazard Risk AI Prediction			
Step	Task	Expected Outcome	Success
1	Tester will login with the account information provided for them	Tester will be taken to the Application information Landing page	
2	Tester will navigate to the Project Dashboard Page and then to Hazard Tab for any project	Tester will navigate to correct page	

3	Tester will press Hazard Prediction Model	Tester will be taken to modal page that displays Hazard Name and Hazard Description	
4	Tester will enter just Hazard Name and press submit	Error as both fields need an input	
5	Tester will enter Description and then submit	A prediction for the Hazard will then be displayed below the fields	
6	Tester will exit the modal page	Will be returned to Hazard Data page	

10.6.7. Document Generation

JAJQ-89 Document Generation			
Step	Task	Expected Outcome	Success
1	Tester will login with the account information provided for them	Tester will be taken to the Application information Landing page	
2	Once all of the tabs in the project have been Completed, navigate to Document Generation > Hazard Identification	Tester will be taken to correct page	
3	Ensure that the query on the page contains all the data expected from the project i.e. Accident Code, associated Hazards and Causes	Query contains expected data	
4	Press Generate Document	A Download will start for a document	
5	Confirm that all the data within the query on the page is also in the document	Document contains correct data	

10.6.8. Data Management

JAJQ-90 Data Management			
Step	Task	Expected Outcome	Success
1	Tester will login with the account information provided for them	Tester will be taken to the Application information Landing page	
2	Navigate to Administration > Lookups	Visible to Tester with Administration Role	

3	Create a new line in the look up tables and navigate to the data tab to verify that this has been added	Added information in the LOV	

10.6.9. Browser Compatibility

JAJQ-81 Browser Compatibility			
Step	Task	Expected Outcome	Success
1	Tester will login with the account information provided for them on a Google Chrome browser using this link: https://oracleapex.com/ords/r/ueacmp/hazard-analysis-application/login	Tester will be taken to the Application information Landing page	
2	Tester will login with the account information provided for them on a Microsoft Edge browser using this link: https://oracleapex.com/ords/r/ueacmp/hazard-analysis-application/login	Tester will be taken to the Application information Landing page	

10.6.10. Page Load Time

JAJQ-92 Page Load Time			
Step	Task	Expected Outcome	Success
1	Tester will login with the account information provided for them	Tester will be taken to the Application information Landing page	
2	Tester will navigate to the Project Dashboard and then will press a navigation to a tab in the child detail interactive grid at the same time as starting a timer. They will then stop this timer once the page has loaded.	Tester will navigate to the tab page in the expected time	
3	This will be repeated 10 times across different tab pages, and an average of all times will be taken.	Average time will be within the expected load time	

10.6.11. User Feedback

JAJQ-82 User Feedback			
Step	Task	Expected Outcome	Success

1	Tester will use the comment icon in the navigation bar to open feedback	Tester will be taken to the feedback page	
2	Tester will write feedback and submit	No errors with writing feedback and submit will take them back to previous page	
3	Navigate to Administration > Feedback > Management Feedback and view the feedback raised changing its status and commenting	Tester will be able to change status and reply to feedback	

10.7. Appendix G (Test Environment)

The test environment below is used to create a baseline for future reference of testing as it includes the exact specification of the software that the tests were performed on. Should there be any changes to requirements or the software that the project uses, the test cases will need running again and a new test configuration will be made.

Project Information		
Application Name	Hazard Analysis Application	
Application Version	Version 1.0	
Release Date	18/05/2026 (Project Portfolio Due Date)	
Requirement Version	Functional_Requirements_V1, Non_Functional_Requirements_V1	
Evidence of Testing	Testing/Formal Testing/Evidence of Testing.xlsx	
Software		
Software Name	**Version**	**Notes**
Windows Operating System	Windows 11 Home – OS Build 26200.7462	Operating system on the device used to develop the application
Oracle APEX	Oracle APEX 24.2.11	Web version of Oracle APEX used for the development of the main Application

Oracle Database	Oracle Database 19c EE Extreme Perf Release 19.0.0.0.0 - Production Version 19.23.0.0.0	Database used to support the implementation of Oracle APEX as well as storing the application data
Google Chrome Browser	144.0.7559.112	Browser used to run Oracle APEX app
Microsoft Edge Browser	144.0.3719.104	Browser used to run Oracle APEX app
Spyder 6	6.0.8	Python IDE used to develop python scripts
Visual Studio Code (User)	1.108.2	IDE used to develop JavaScript scripts
Terminal	1.23.20211.0	Device Terminal used to run scripts

10.8. Appendix H (Table Definition)

The table below provides a description of each table and its purpose within the system.

Table Name	**Description**
H_ACCIDENT_DATA	Contains all data for the accident tab
H_ACCIDENT_TYPE	Is a table used to populate LOVs to determine an Accidents Type. FK in H_ACCIDENT_DATA
H_CAUSE	Contains all data for the cause tab
H_CONTROL	Contains all data for the control tab
H_HAZARD	Contains all data for the hazard tab
H_HAZARD_SOURCE	Is a table used to populate LOVs to determine the source of a Hazard. FK in H_HAZARD.
H_HAZARD_TYPE	Is a table used to populate LOVs to determine and Hazards Type. FK in H_HAZARD
H_LINK_ACCIDENT_CAUSE	Link table which allows for accidents to be paired with causes. FKs for both Accident and Cause tables.
H_LINK_ACCIDENT_CONTROL	Link table which allows for accidents to be paired with controls. FKs for both Accident and Control tables.
H_LINK_ACCIDENT_HAZARD	Link table which allows for accidents to be paired with hazards. FKs for both Accident and Hazard tables.
H_LINK_CAUSE_CONTROL	Link table which allows for causes to be paired with controls. FKs for both Cause and Control tables.

H_LINK_CONTROL_REFERENCE	Link table which allows for controls to be paired with references. FKs for both Control and Reference tables.
H_LINK_HAZARD_CAUSE	Link table which allows for hazards to be paired with causes. FKs for both Hazard and Cause tables.
H_LINK_HAZARD_CONTROL	Link table which allows for hazards to be paired with controls. FKs for both Hazard and Control tables.
H_LOCATION	Is a table used to populate LOVs to determine the location of where data entry will be implemented. FK in H_ACCIDENT_DATA and H_HAZARD.
H_PROJECT	Contains all information related to a Project, it is used as an FK in most tables to hide information to users that aren't associated with the Project.
H_PROJECT_TAB	A table that tracks that data and link tabs that are assigned to a Project.
H_PROJECT_TAB_STATUS	A table used to populate LOVs and keep a single source for a tabs status.
H_REFERENCE	Contains all data for the reference tab
H_REFERENCE_TYPE	Is a table used to populate LOVs to determine a Refernces Type. FK in H_REFERENCE
H_STATUS	Used to determine the status of a line in H_ACCIDENT_DATA, H_CAUSE, H_CONTROL, H_HAZARD and H_REFERENCE
H_TAB	A table containing the Types of tabs that are required to be filled out on a project
H_TEAM	Table that contains information regarding a team so that Users can be assigned to them via PROJECT_TEAM_MEMBERS
H_USER_ROLES	Used to assign User Roles to a Username.
PROJECT_TEAM_MEMBERS	Used to assign Users Username to a Project_ID to ensure Users can only see data from projects they are assigned to.